INFN

POLITECNICO MILANO 1863

DIPARTIMENTO DI ELETTRONICA INFORMAZIONE E BIOINGEGNERIA

RadLab

Laboratory of Radiation Detectors and Nuclear Electronics

Topics for Master Thesis: (for ELN, BIO, NUC, FIS,.. students)

- Sensors, electronics and instrumentation for X and gamma rays applications in medical diagnostics, fundamental physics, astronomy, study of matter and industrial applications
- Integrated circuits for signal processing of sensors signals in scientific and industrial applications

Pre-requisites:

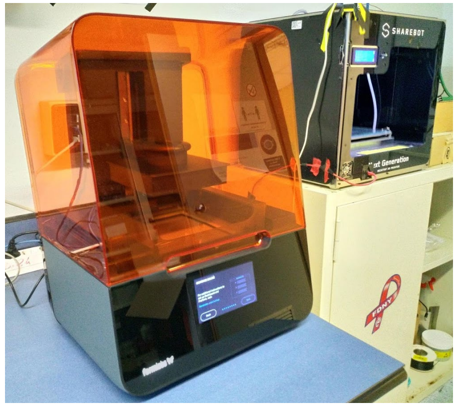
- interest to experimental activity in laboratory
- good background in electronics (for thesis in integrated circuits design)
- motivation, curiosity.....all the rest you will learn in the lab...!

The RadLab Group

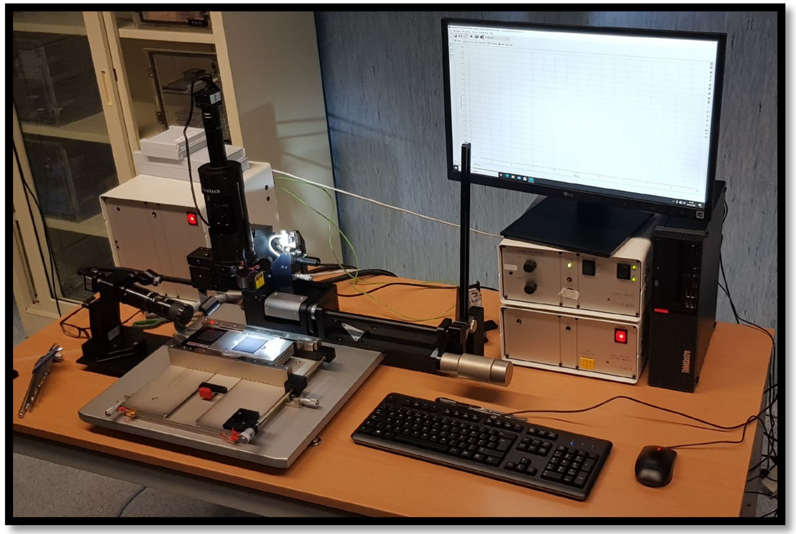
- 3 staff, 12 Ph.D. students and post-docs, 20 M.Sc. thesis students/year
- Cadence workstations, Electronics assembly and test instrumentation, Wire bonding and Flip Chip, 3D printers, vacuum and climatic chambers, X/γ photon sources,…

Wedge bonder

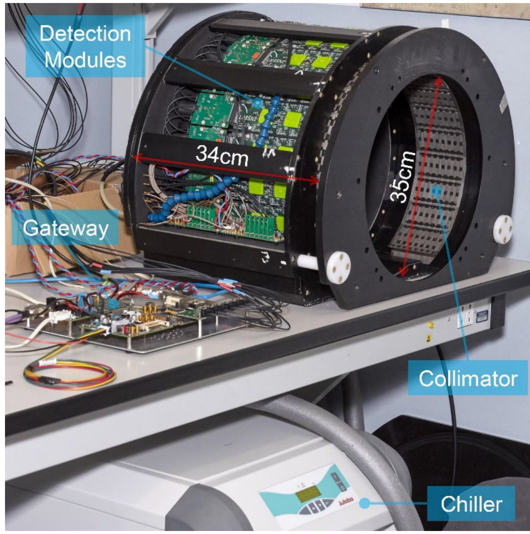
3D printer



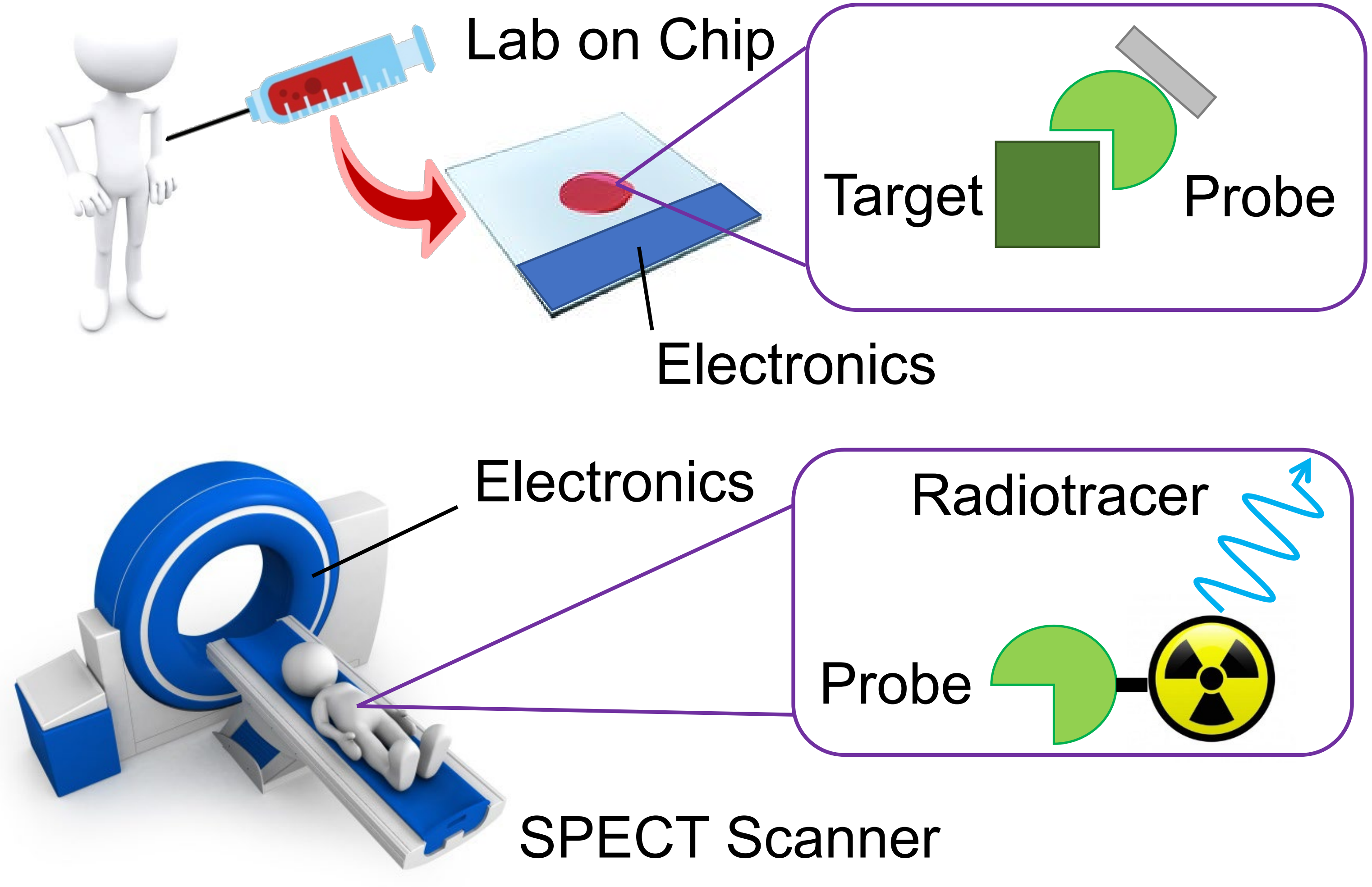
Flip-chip



Instrumentation



Diagnostics



Development of detectors, electronics and instrumentation for innovative Medical Imaging techniques

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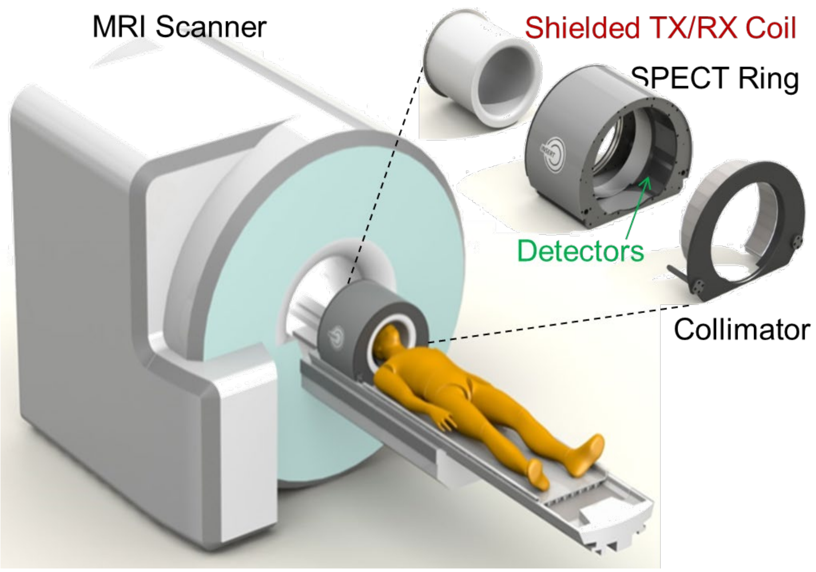
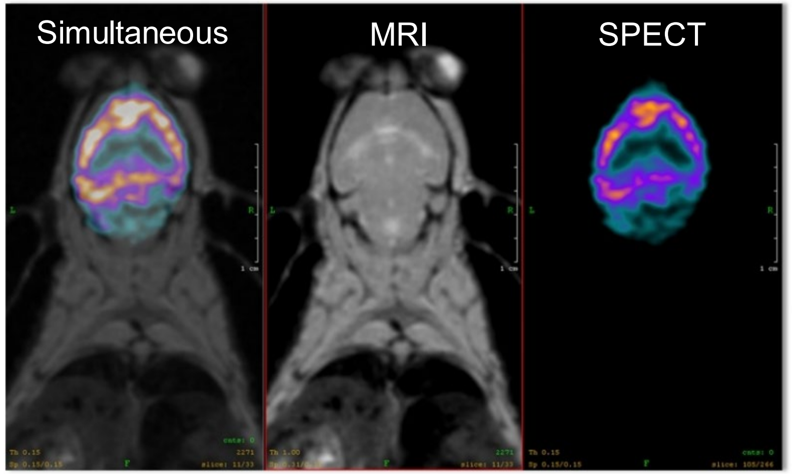
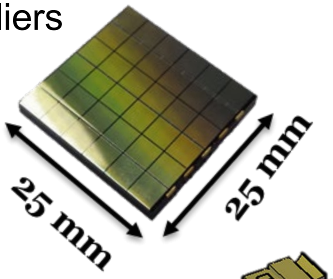
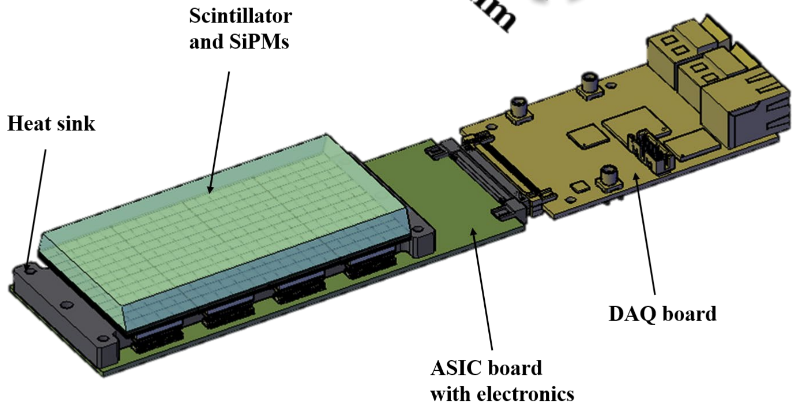
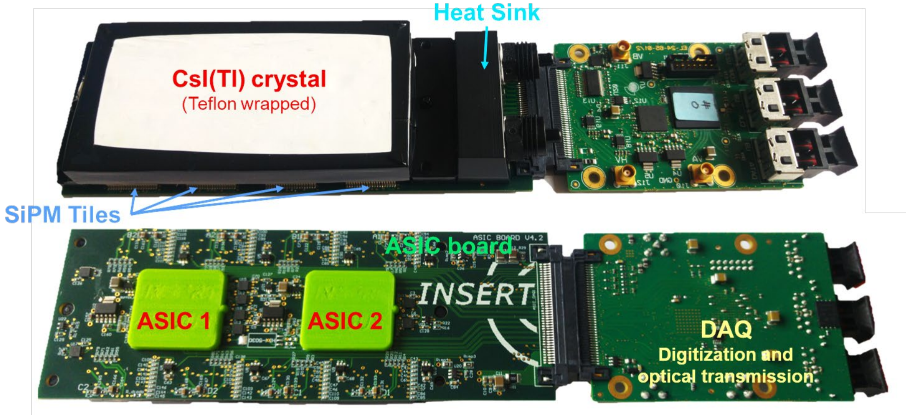


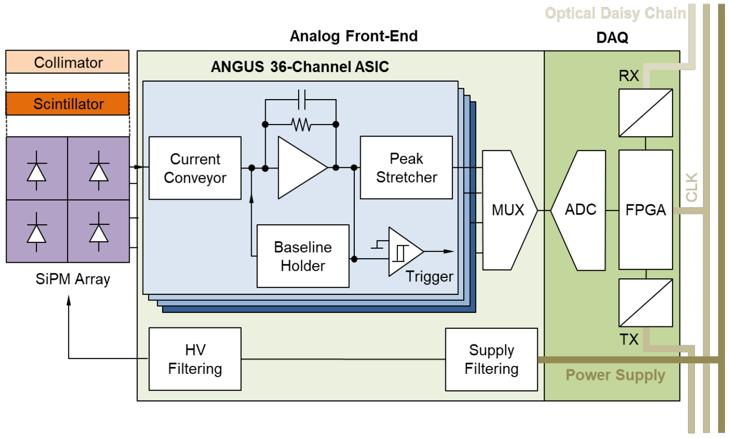


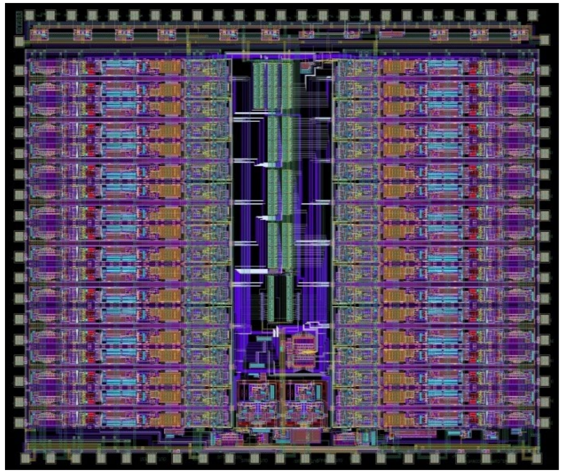
Silicon Photomultipliers (SiPMs)



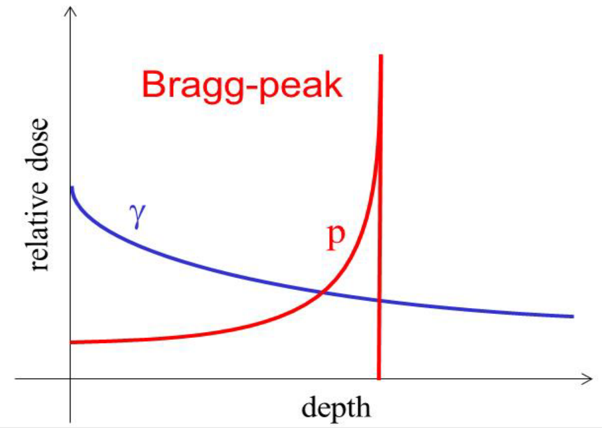




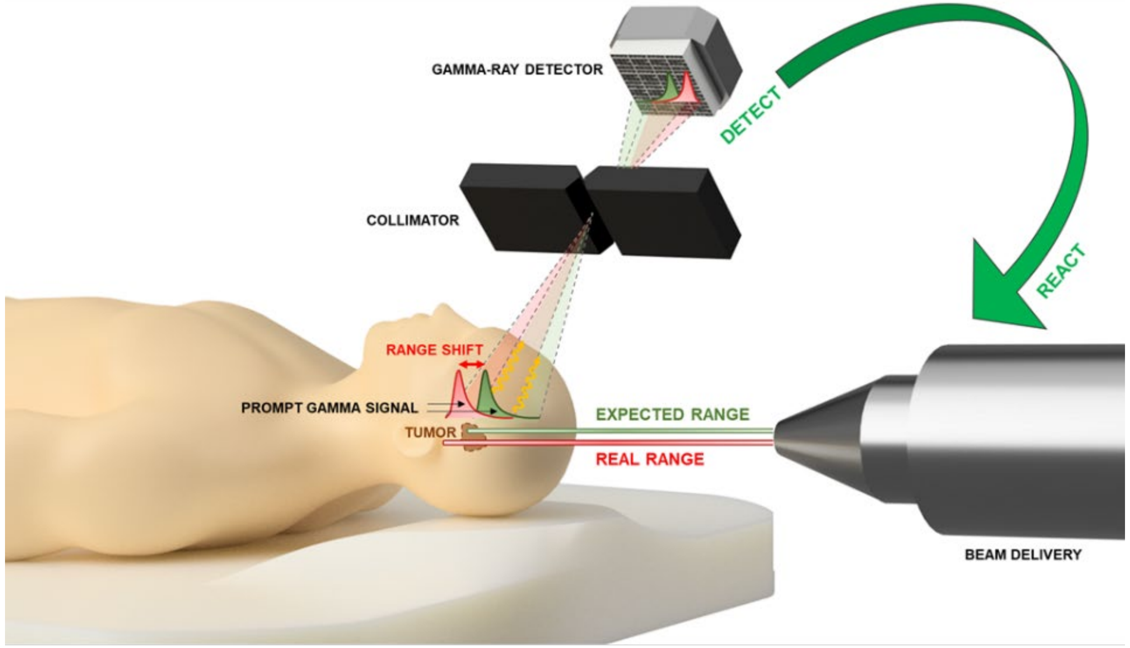


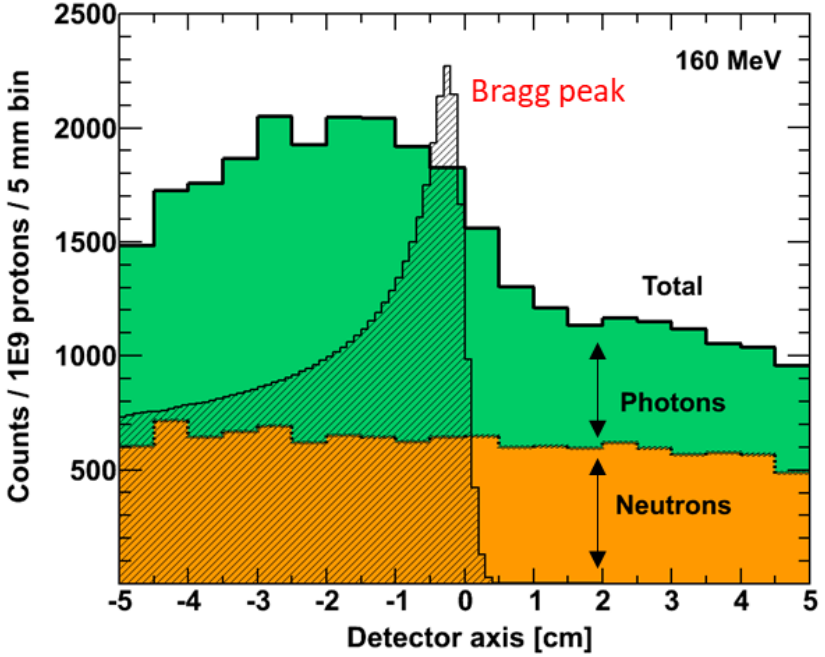


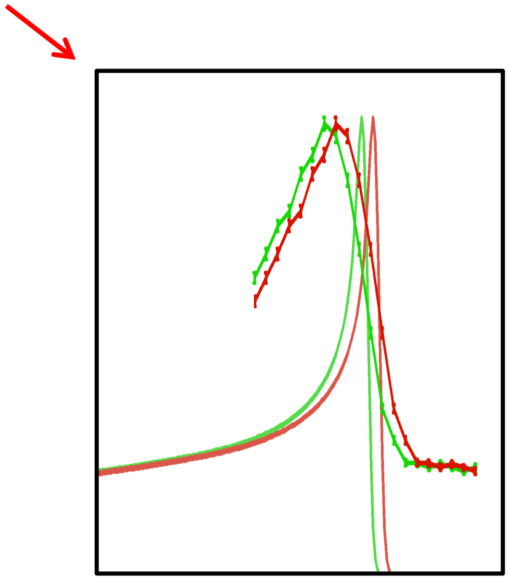
Processing ASIC (amplitude, timing)



The measurement of the proton beam range in the target is very important: real range of proton beams in patients may contain uncertainties of up to 10-15 mm (uncertainties on tissue composition, density, organ motions, patient positioning, etc).







Detection systems for Particle Therapy (2)

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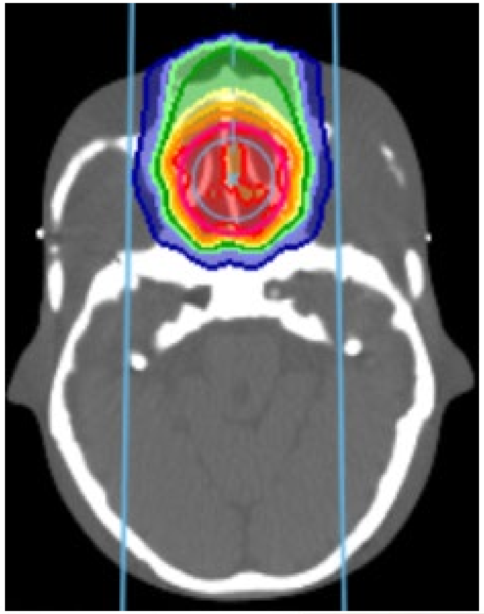
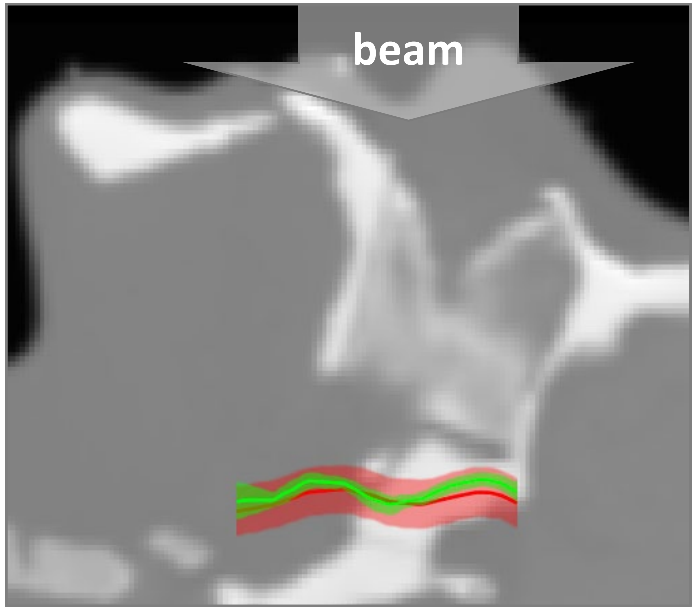
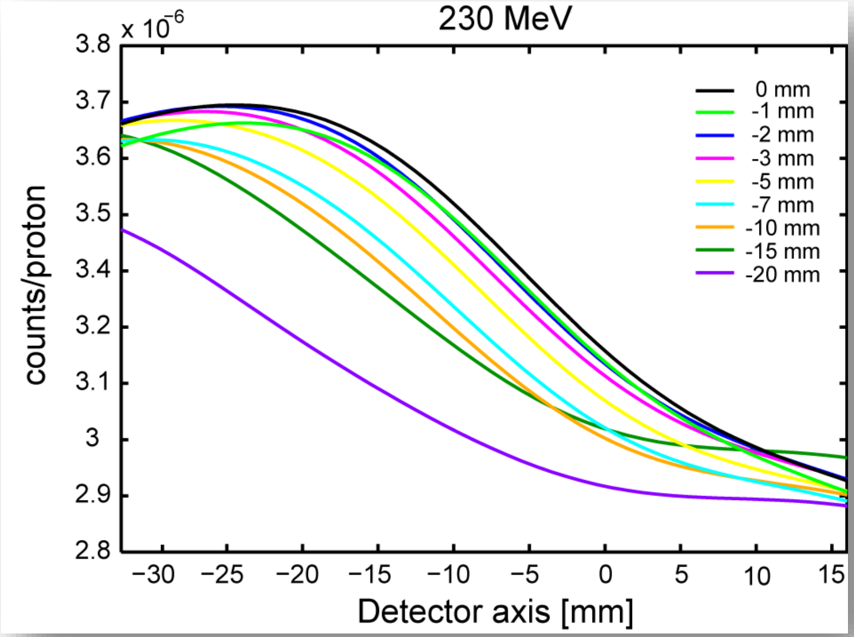
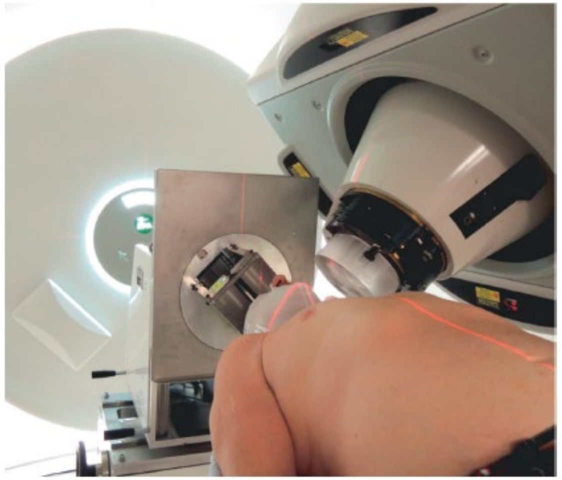






Shift measurements



Planning uncertainty **> 5 mm**
(margin of 3.5% + 2 mm)
Measurement uncertainty (1.5σ)
≈ 2.0 mm

Clinical trial of PG camera for Proton irradiation

Development of PG camera for Carbon irradiation (CNAO):

- Two orders of magnitude less carbon ions than protons used for irradiation (issue partially compensated by higher PG yield of carbon vs. proton)
- Secondary gammas reduces the range-end falloff
- Higher neutron background (vs. proton irradiation)

Detectors and electronics development

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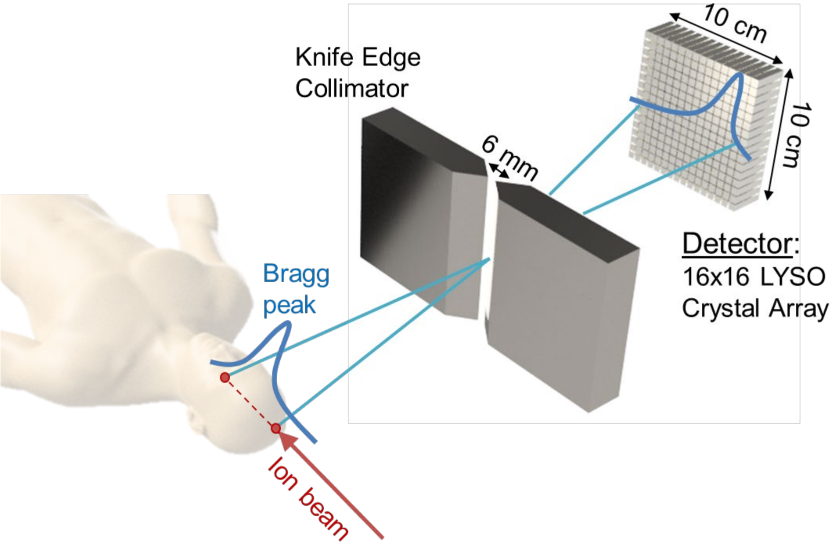
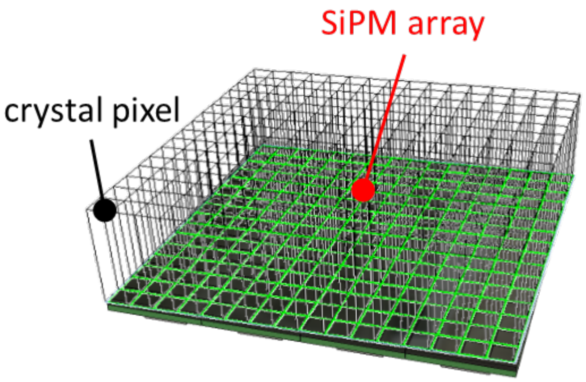
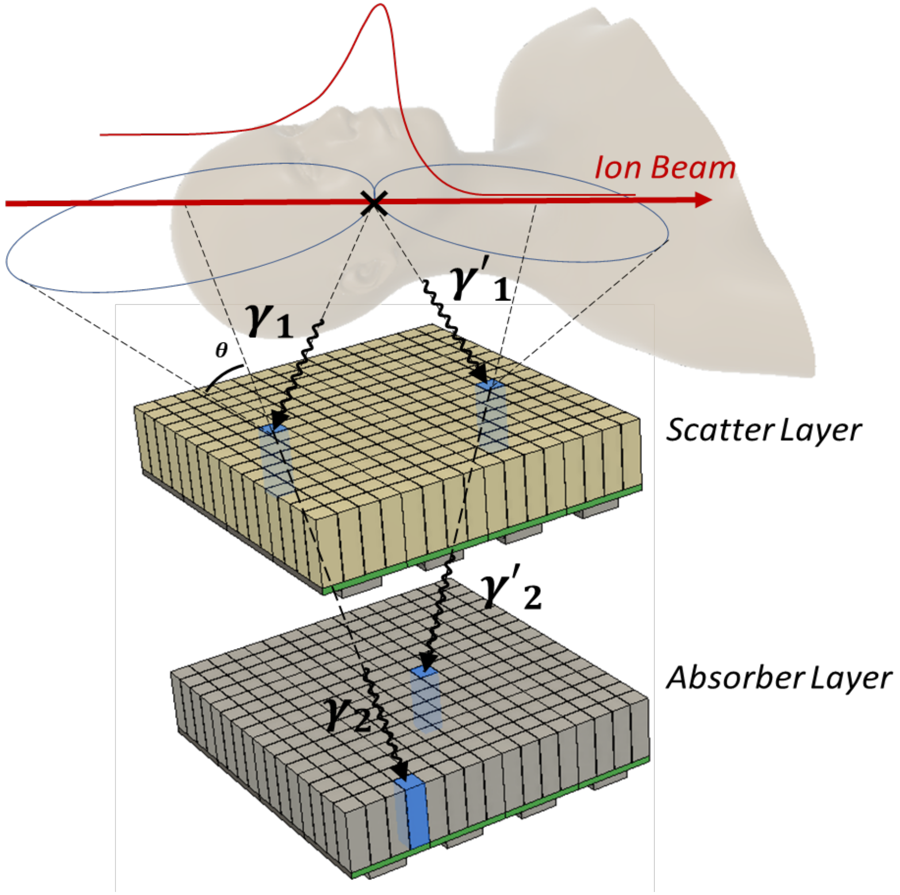
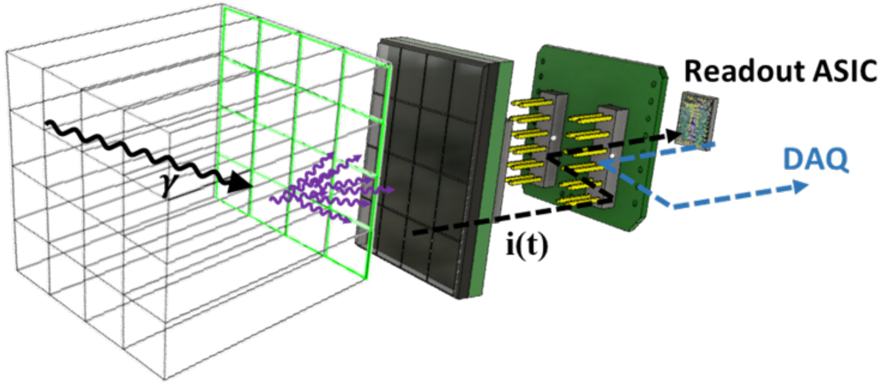


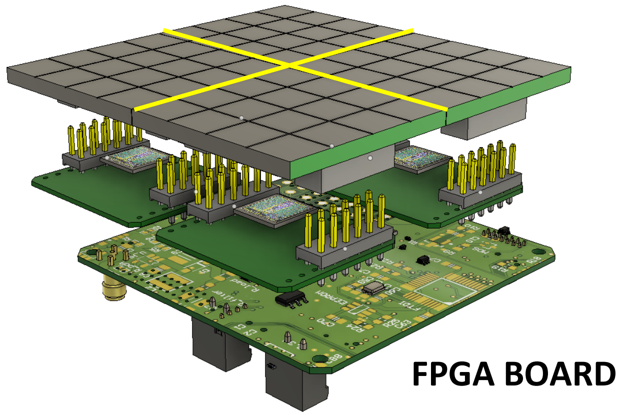




16–SIPM/ASIC UNIT:



64–SIPM UNIT:



Compton – Camera (collimator-less)

Gamma-Neutron discrimination

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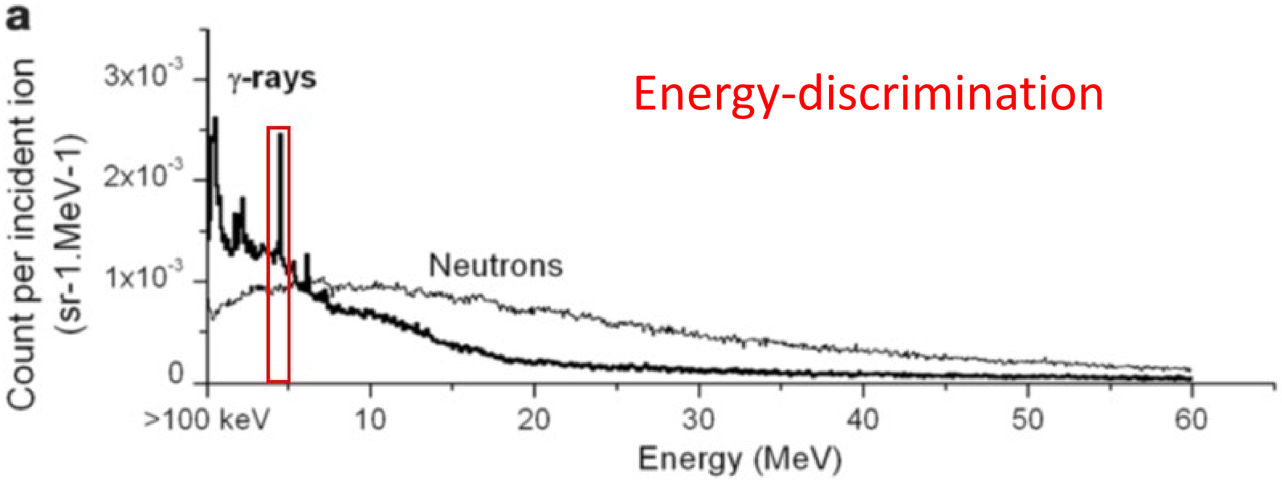
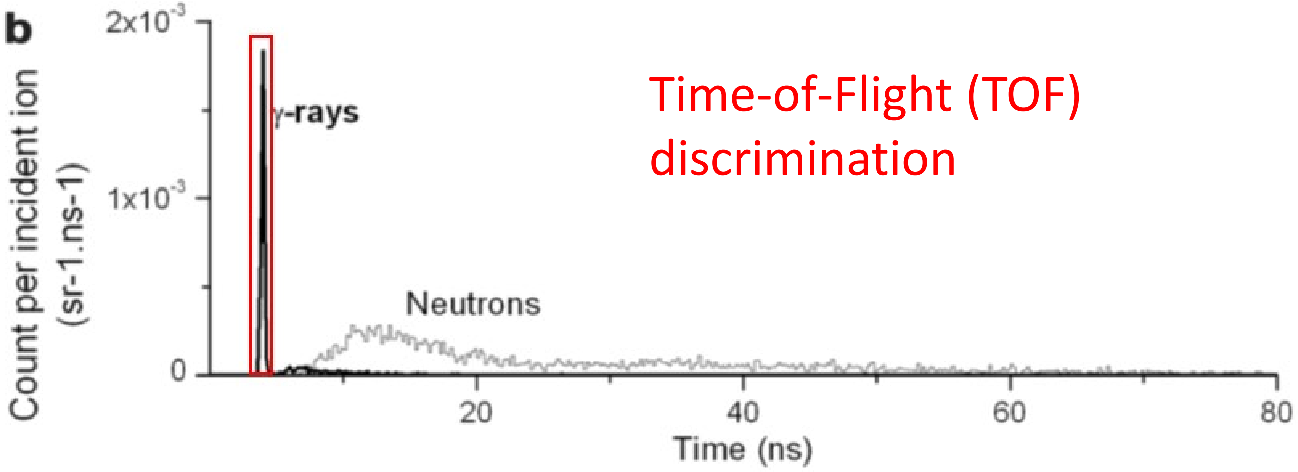
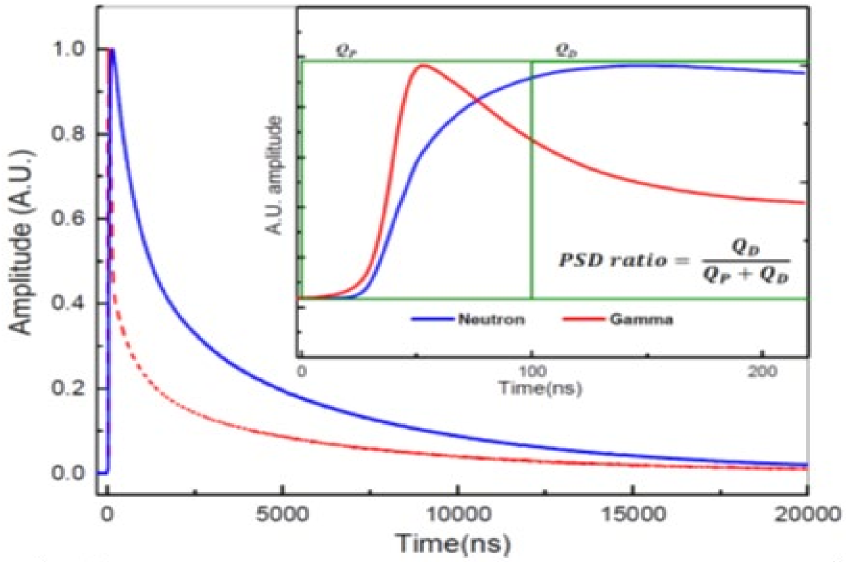
Energy-discrimination

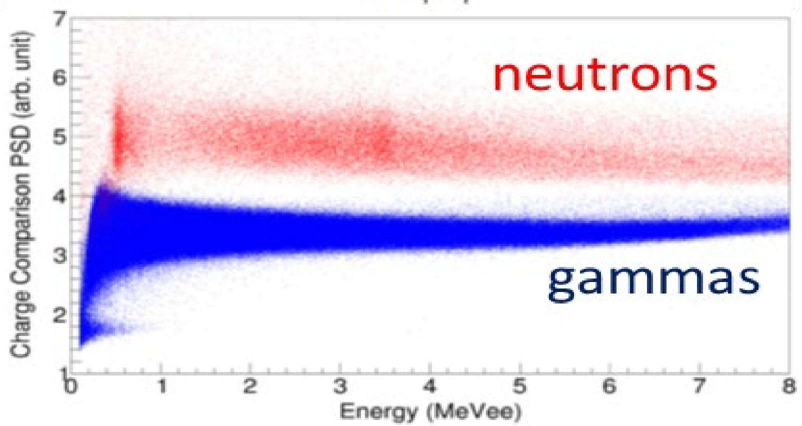


Time-of-Flight (TOF) discrimination



Pulse-shape-discrimination (PSD)





Boron Neutron Capture Therapy

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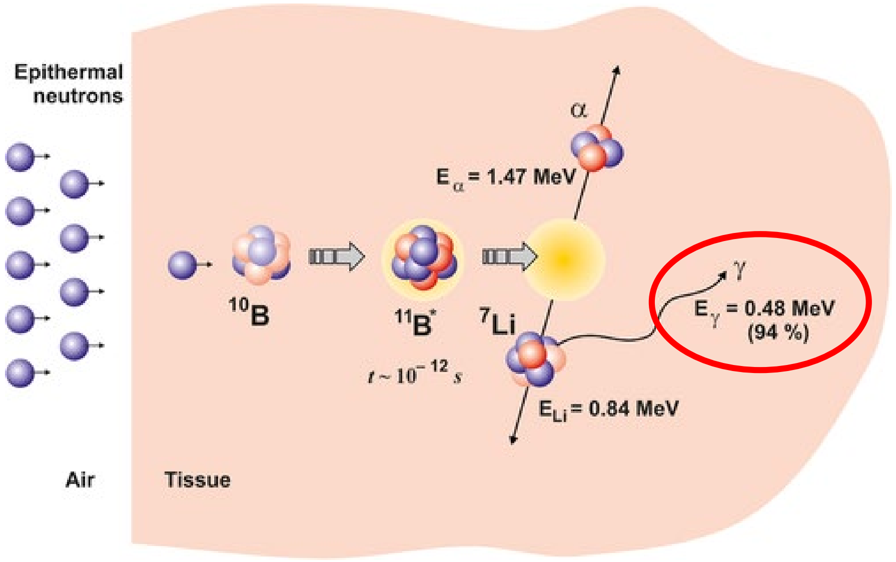


Neutron Capture Agents

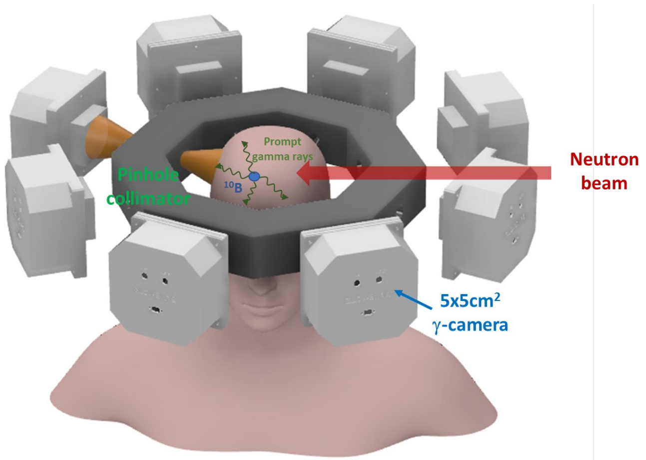
^{10}B-borono-L-phenylalanene

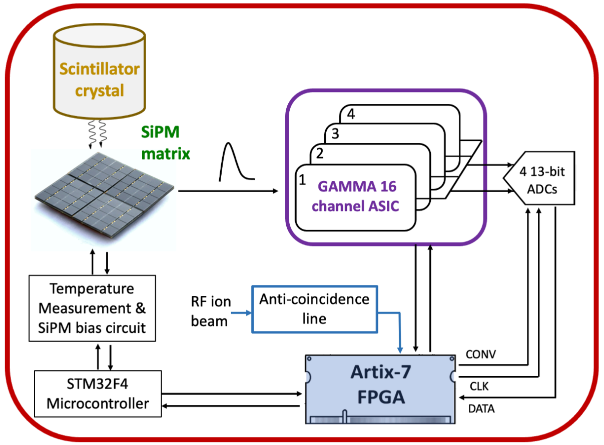
- Thermal neutron cross-section of ^{10}B is 3838 barns
- Neutron capture results in release of damaging high-LET alpha particles and ^{7}Li nucleus:

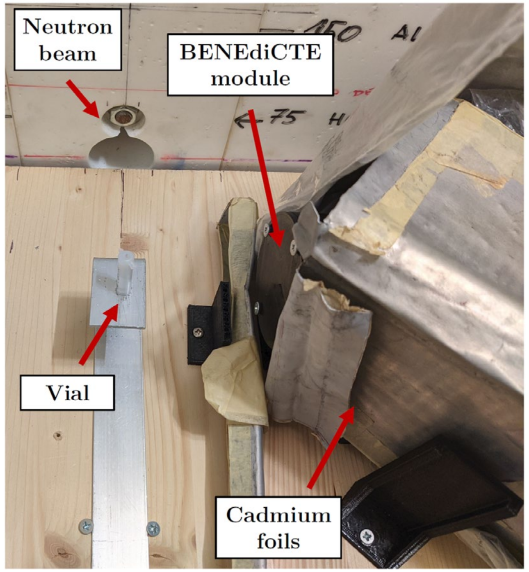
$$^{10}B + n_{th} \rightarrow [^{11}B]^{*} \rightarrow \alpha + {}^{7}Li + \gamma\ (0.48\ \&\ 2.31\ \text{MeV})$$

- BPA is preferentially absorbed by cancer cells & is **clinically approved for use in neutron capture therapy** in Japan and elsewhere

Goal: SPECT for BNCT







Machine Learning for event reconstruction in medical imaging

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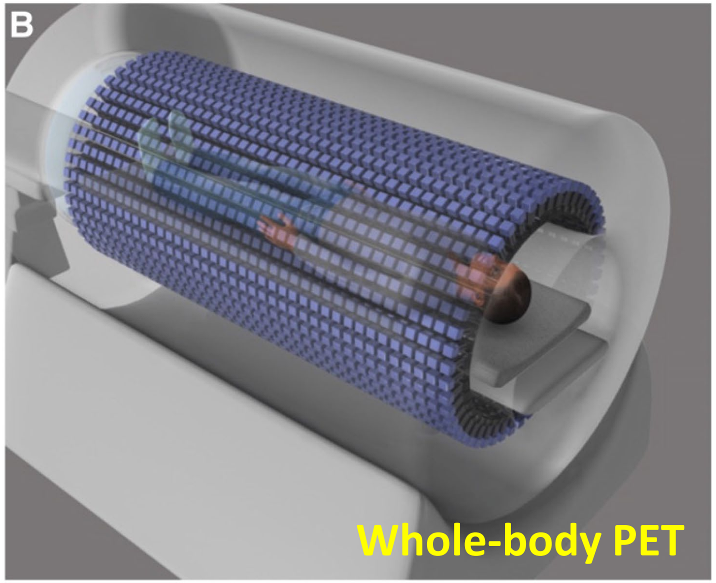
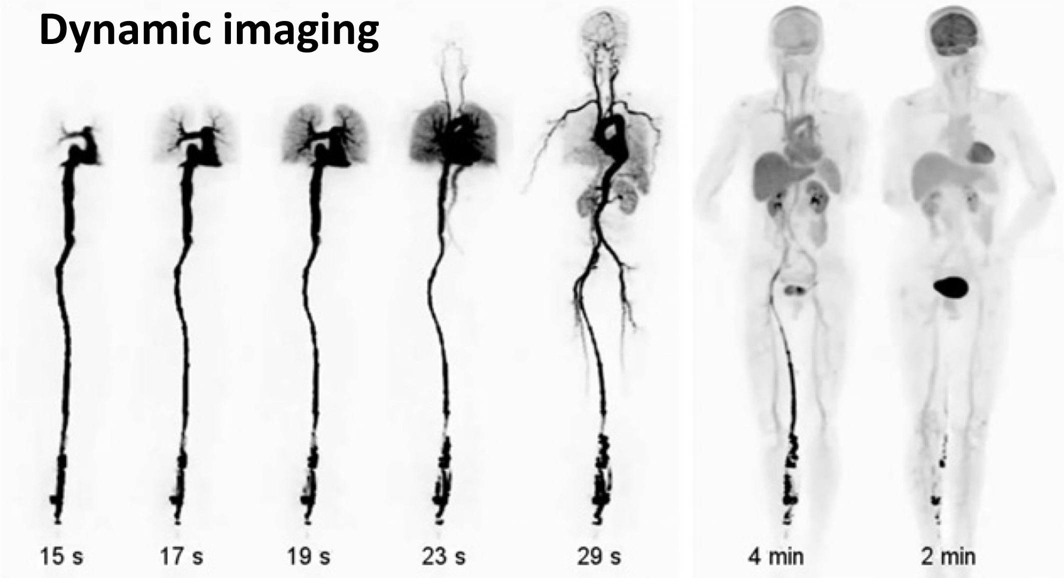
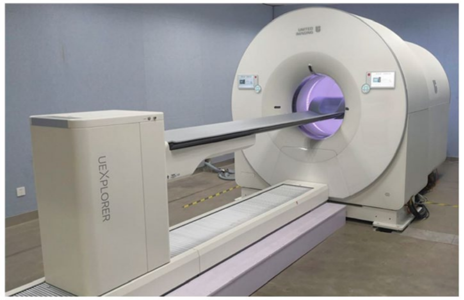




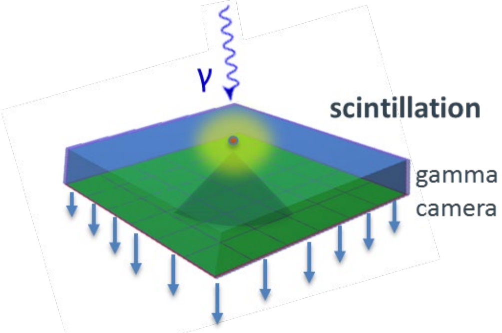
Motivations for ML:
Embedded event reconstruction in very large scanners, multi-modality systems, portable systems...



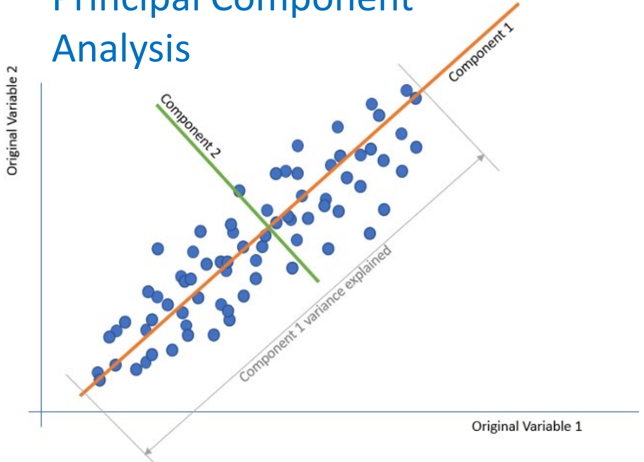
Explorer (UC-DAVIS)

Approaches:

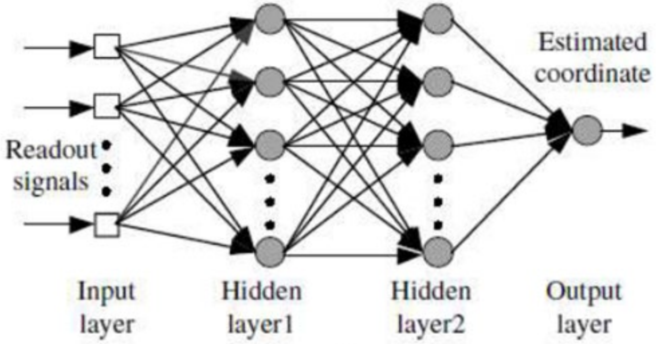
- Multiplexing (e.g. PCA)
- Machine Learning



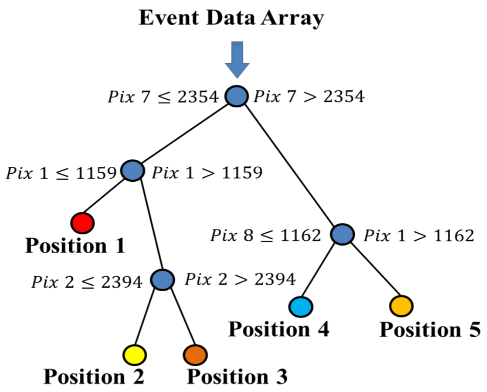
Principal Component Analysis



Neural Network



Decision Tree



ANNA (Analog Neural Network ASIC)

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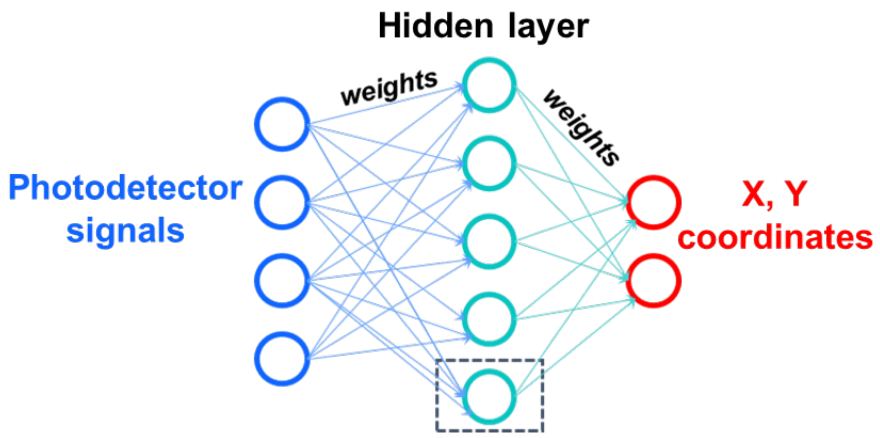
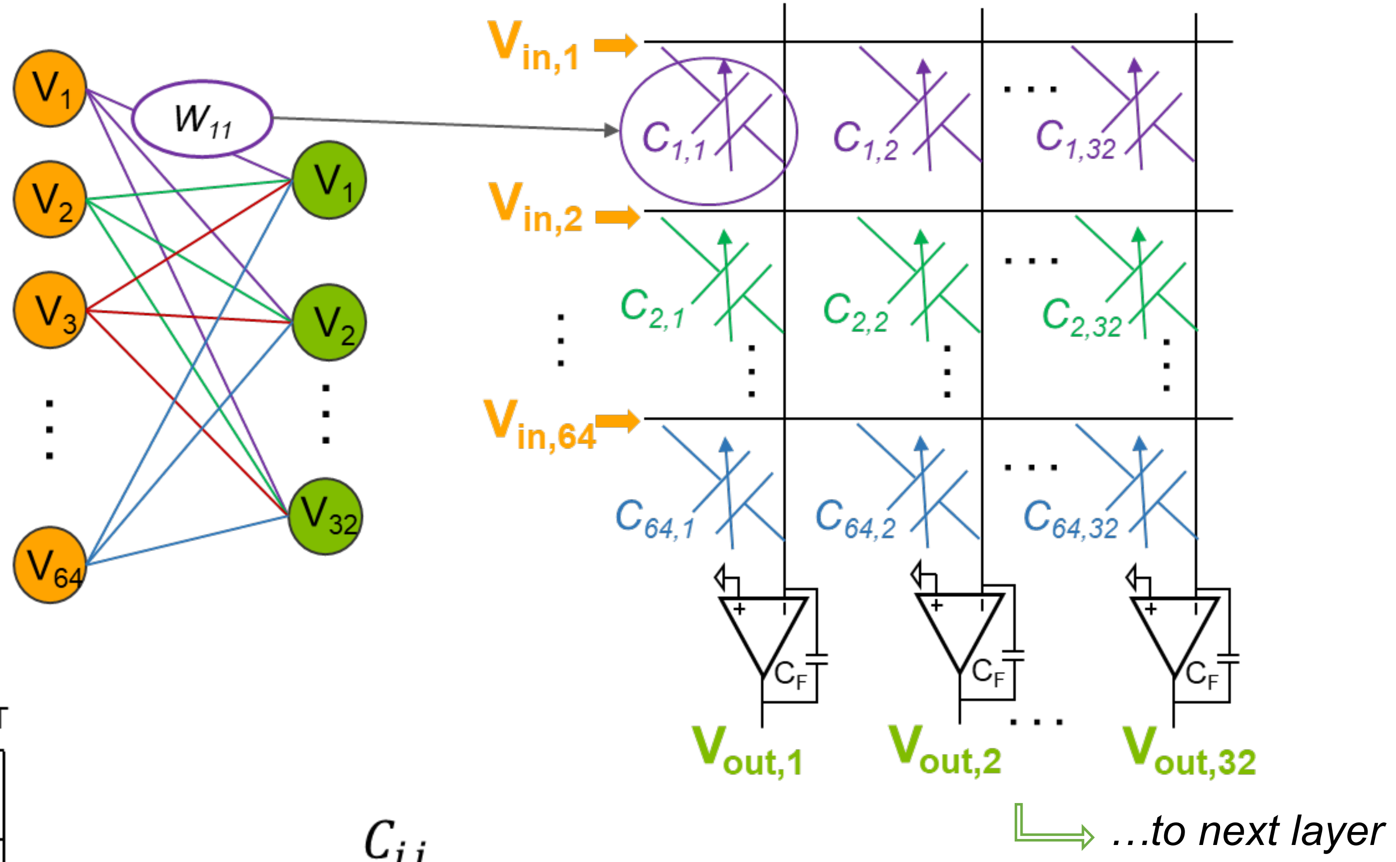
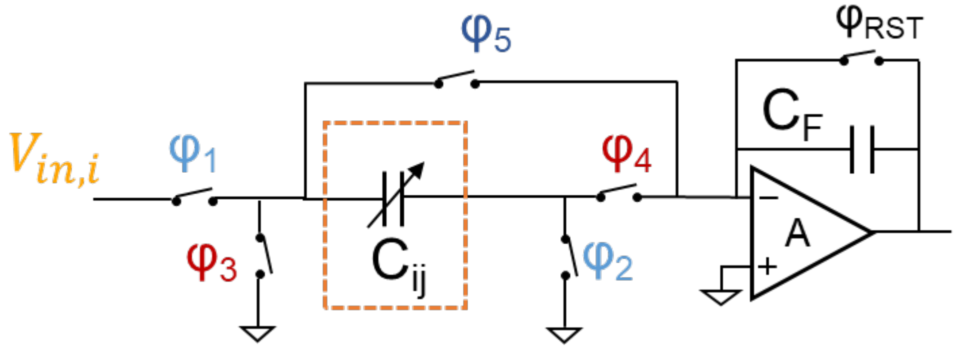






$$V_{out,j} = V_{in,i}\frac{C_{ij}}{C_F}$$

LUXOTTICA

RadLab

Joint Research Center with LUXOTTICA Smart Eyewear Lab

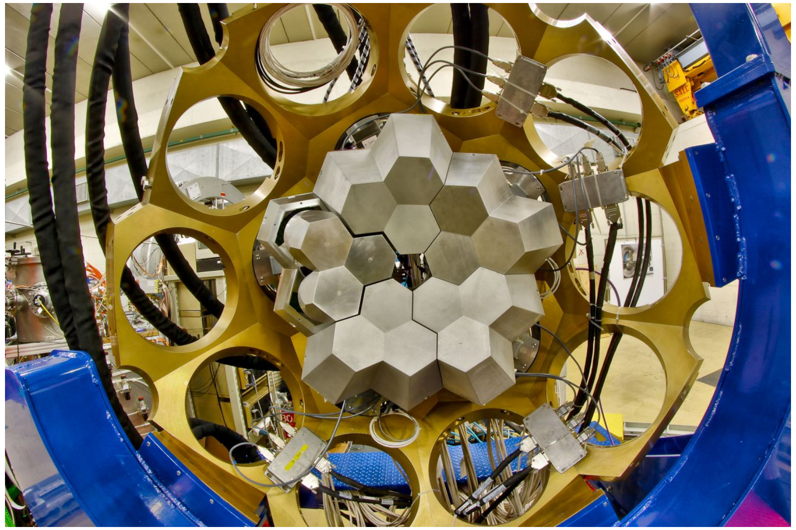
Goals:

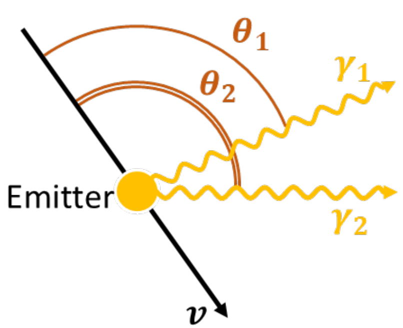
- Augmented Reality
- Vitals monitoring
- Diagnostics

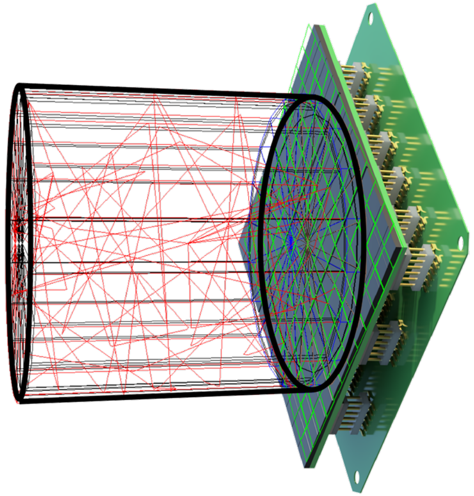
Key Technologies:

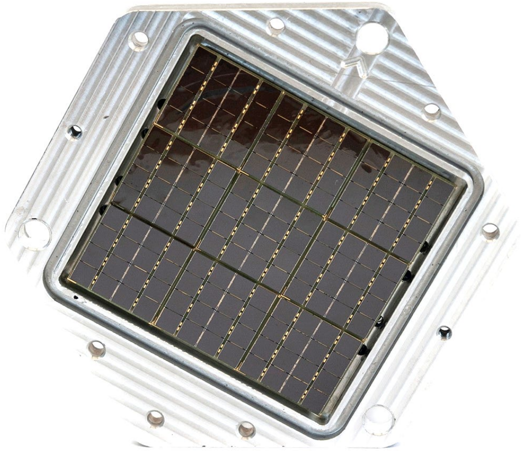
- Metasurfaces, lens hologram
- Sensors and Cameras
- Eye Tracking
- Embedded Machine Learning
- Ultra low power
- Microcontrollers
- Optoelectronics

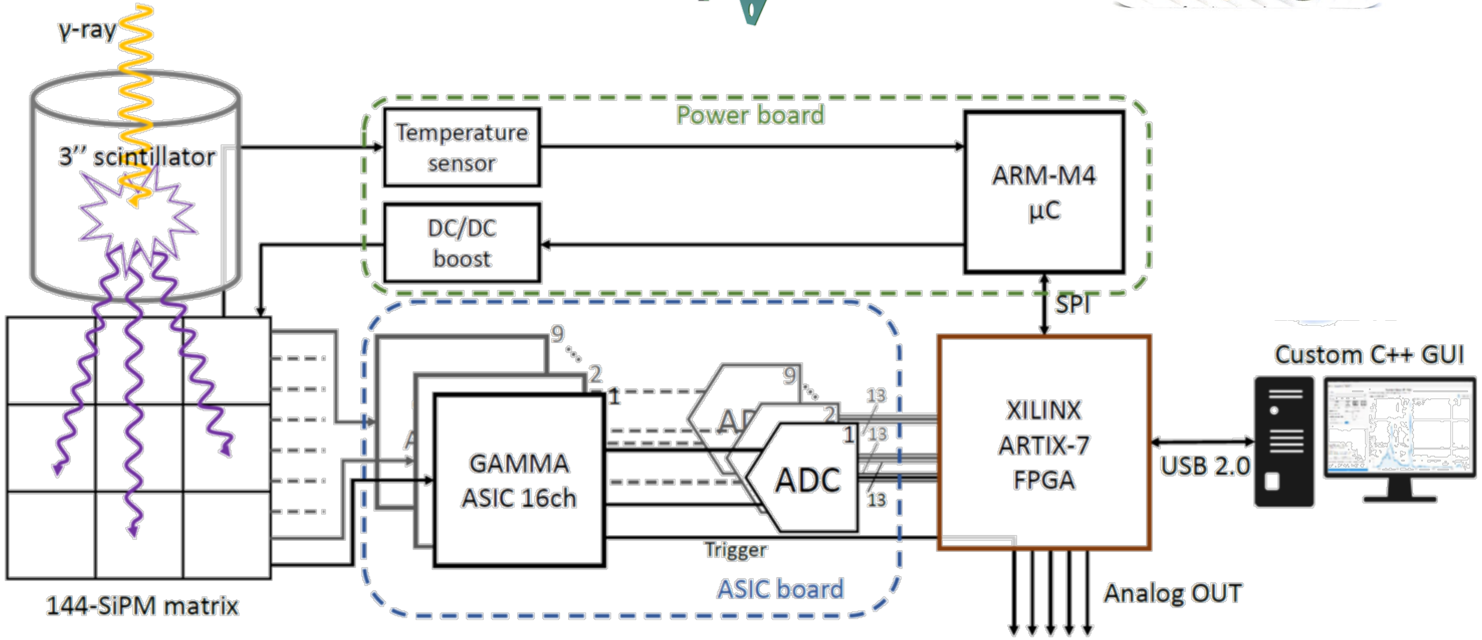
Electronics for fundamental physics experiments



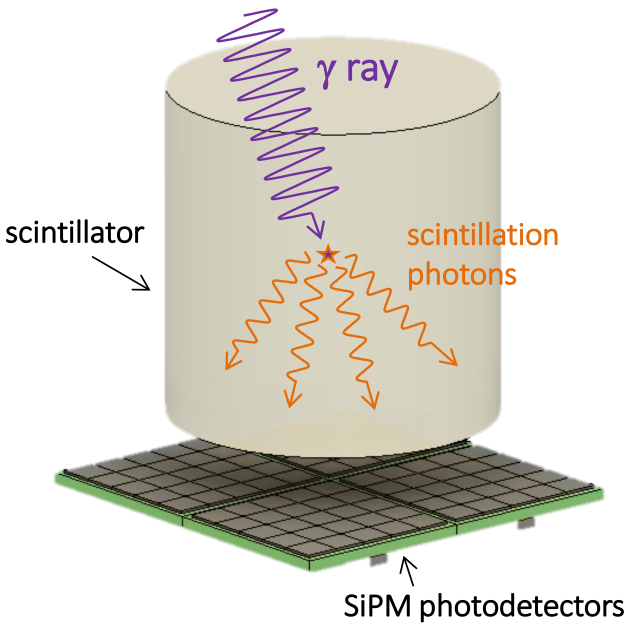






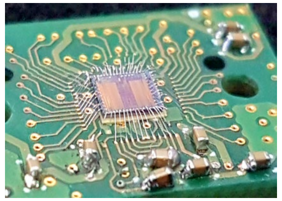


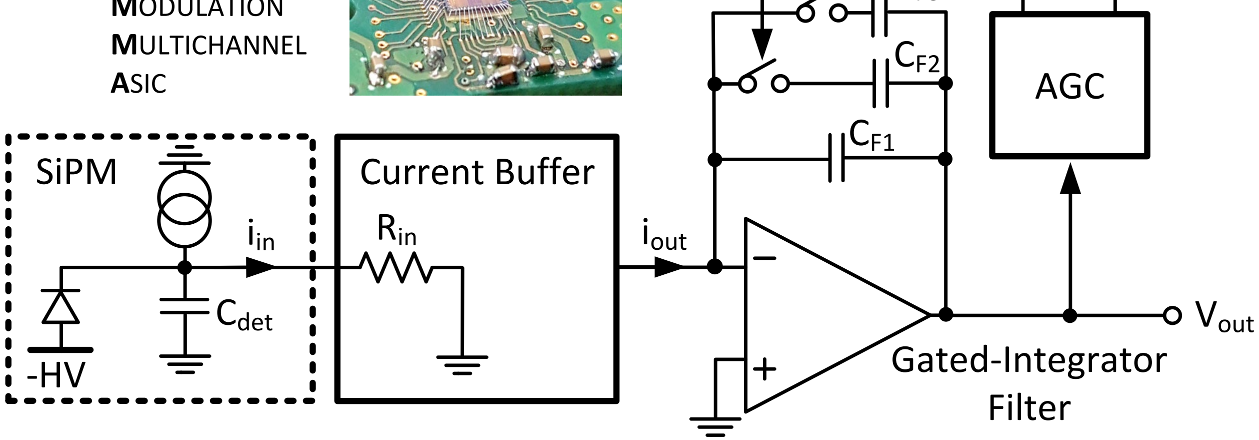
Automatic Gain Control (AGC) ASIC to cover a high dynamic range (1ph → ~ 10.000ph) in γ-spectroscopy



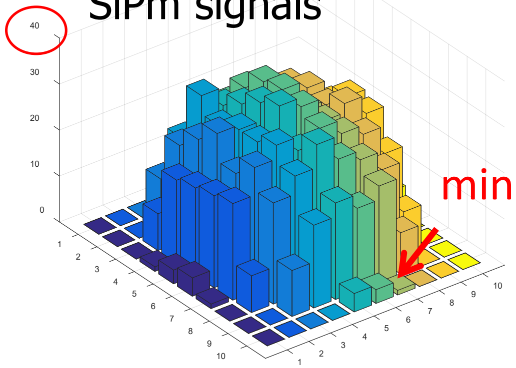
GAMMA project (Spectroscopy and Imaging of wide-range gamma rays), supported by Italian INFN

GAIN
AMPLITUDE
MODULATION
MULTICHANNEL
ASIC

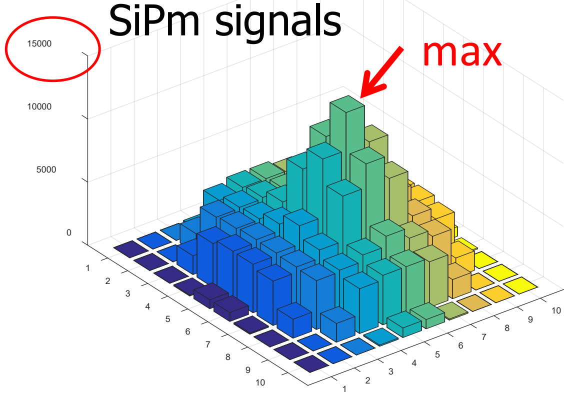


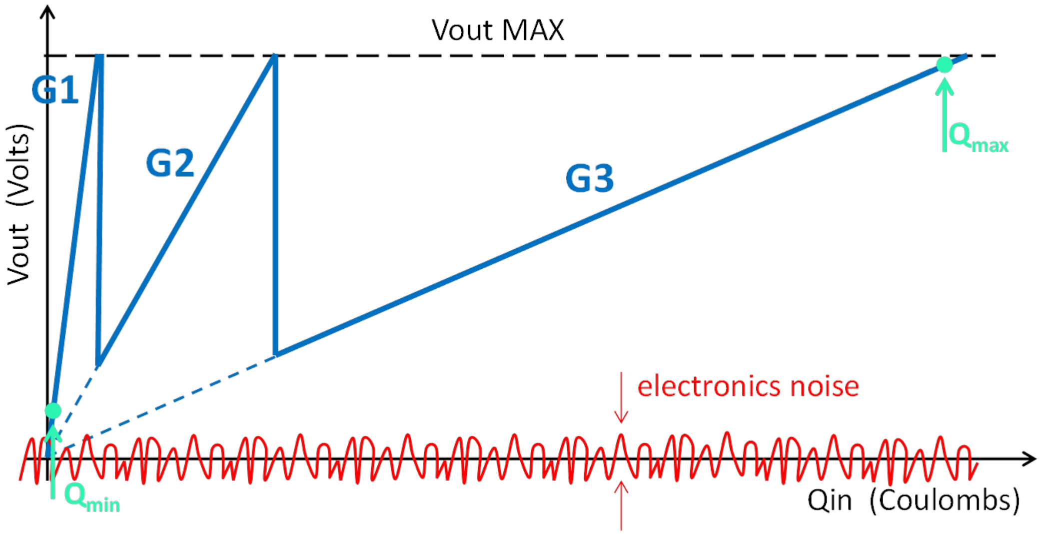


200keV gamma ray:
SiPm signals



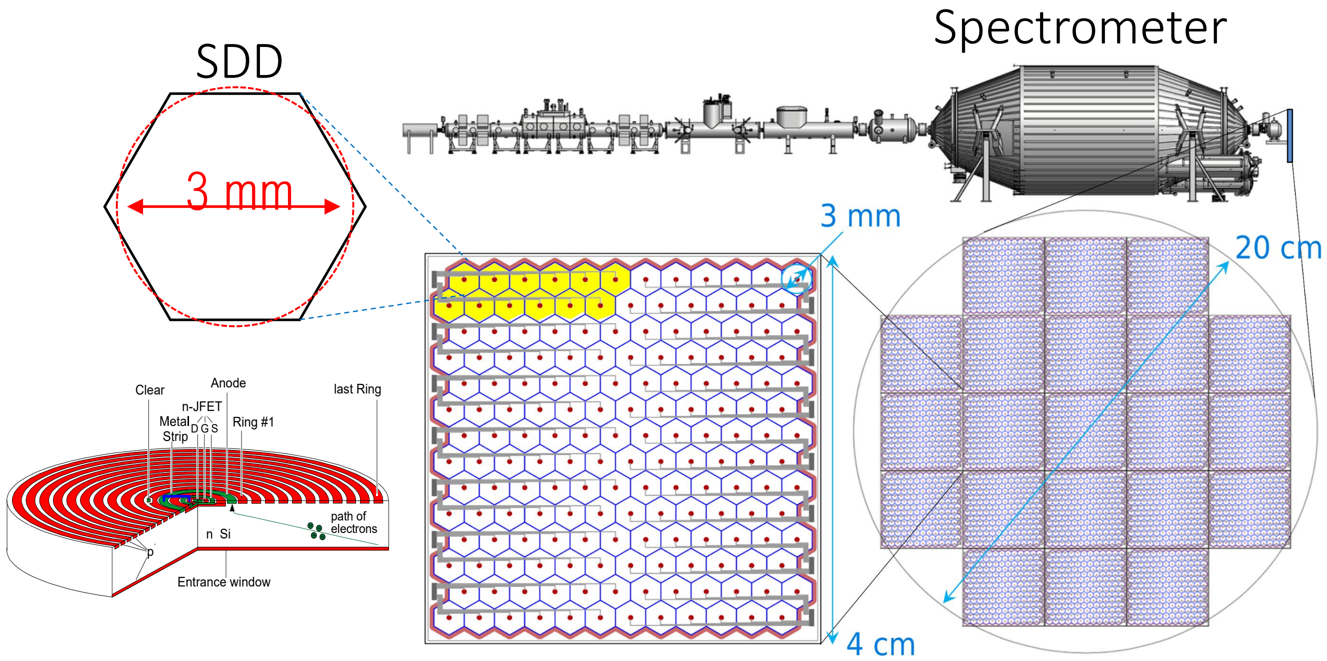
20MeV gamma ray:
SiPm signals

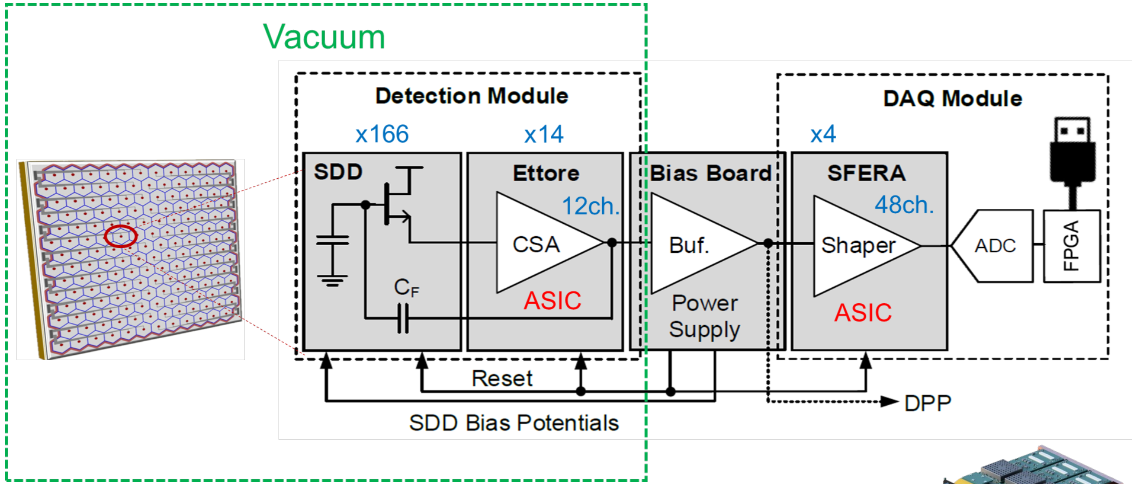




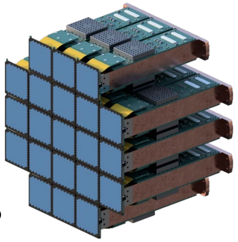
The TRISTAN Detector

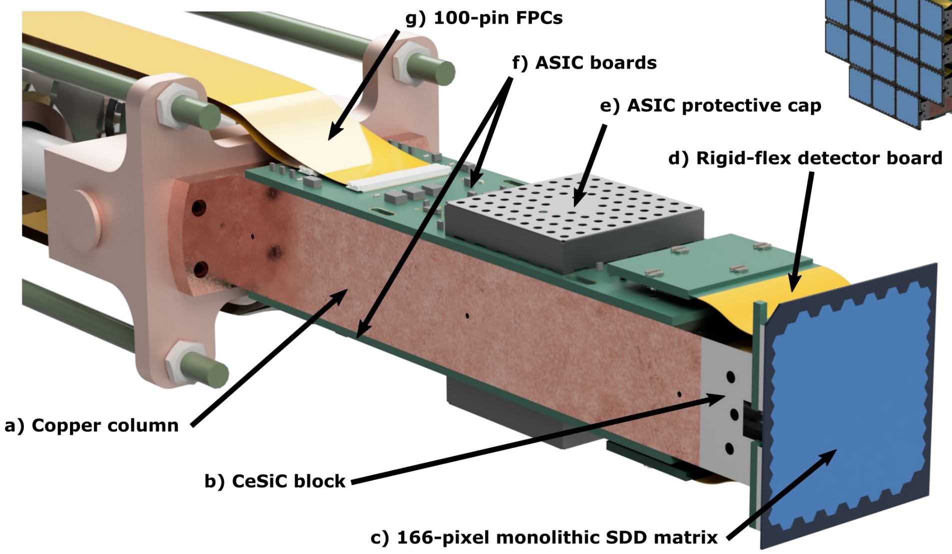
RadLab

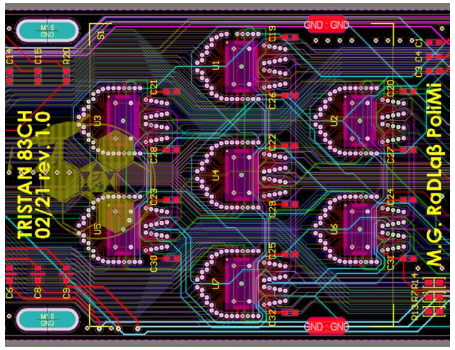


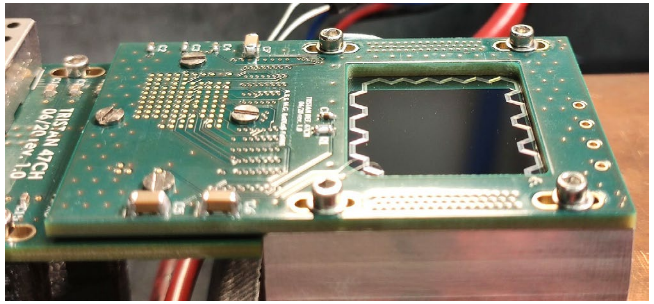


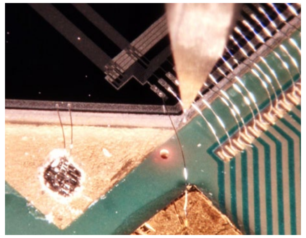
- ~3500 total channels
- 100kcps per channel





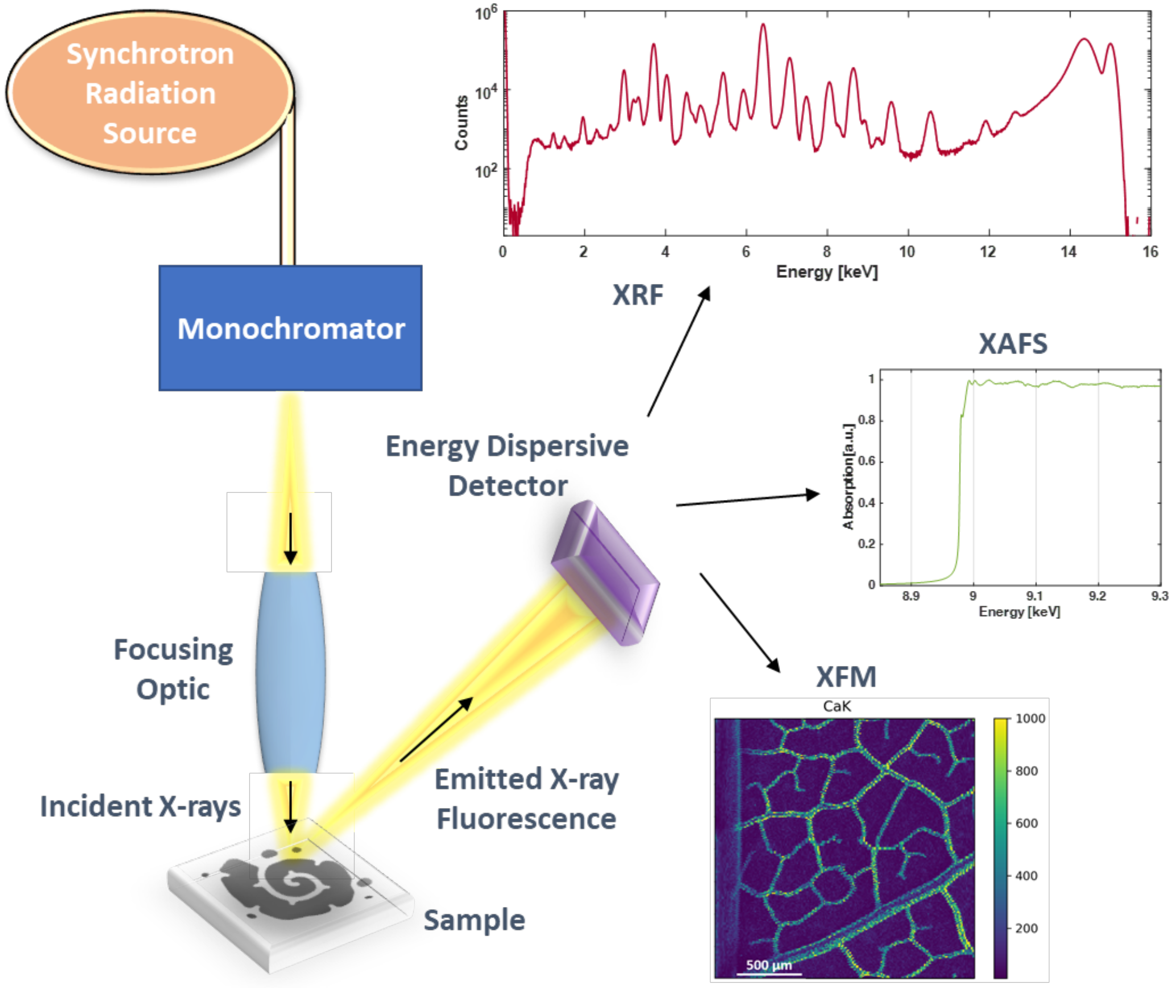


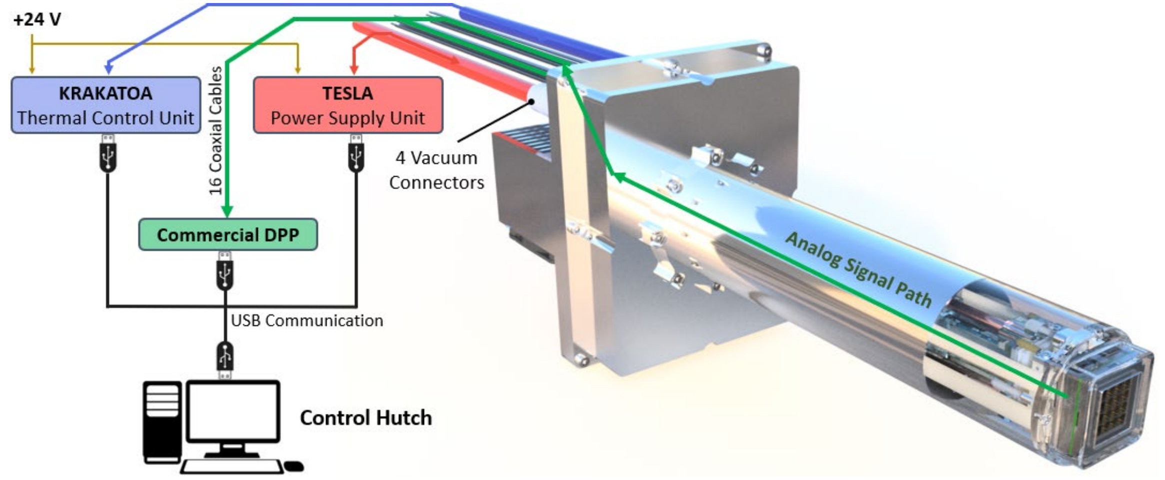


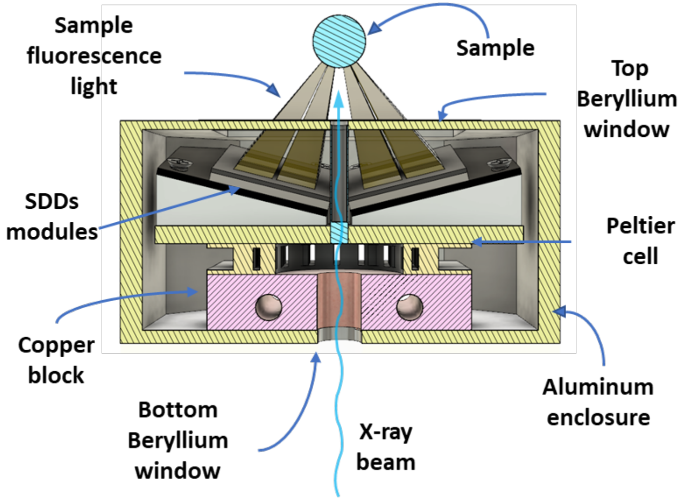


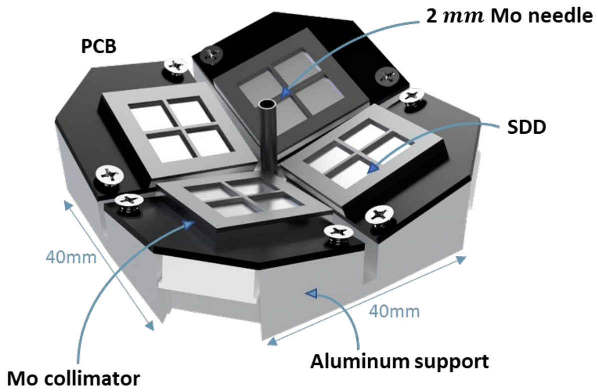
High-Throughput X-ray Spectrometers

RadLab

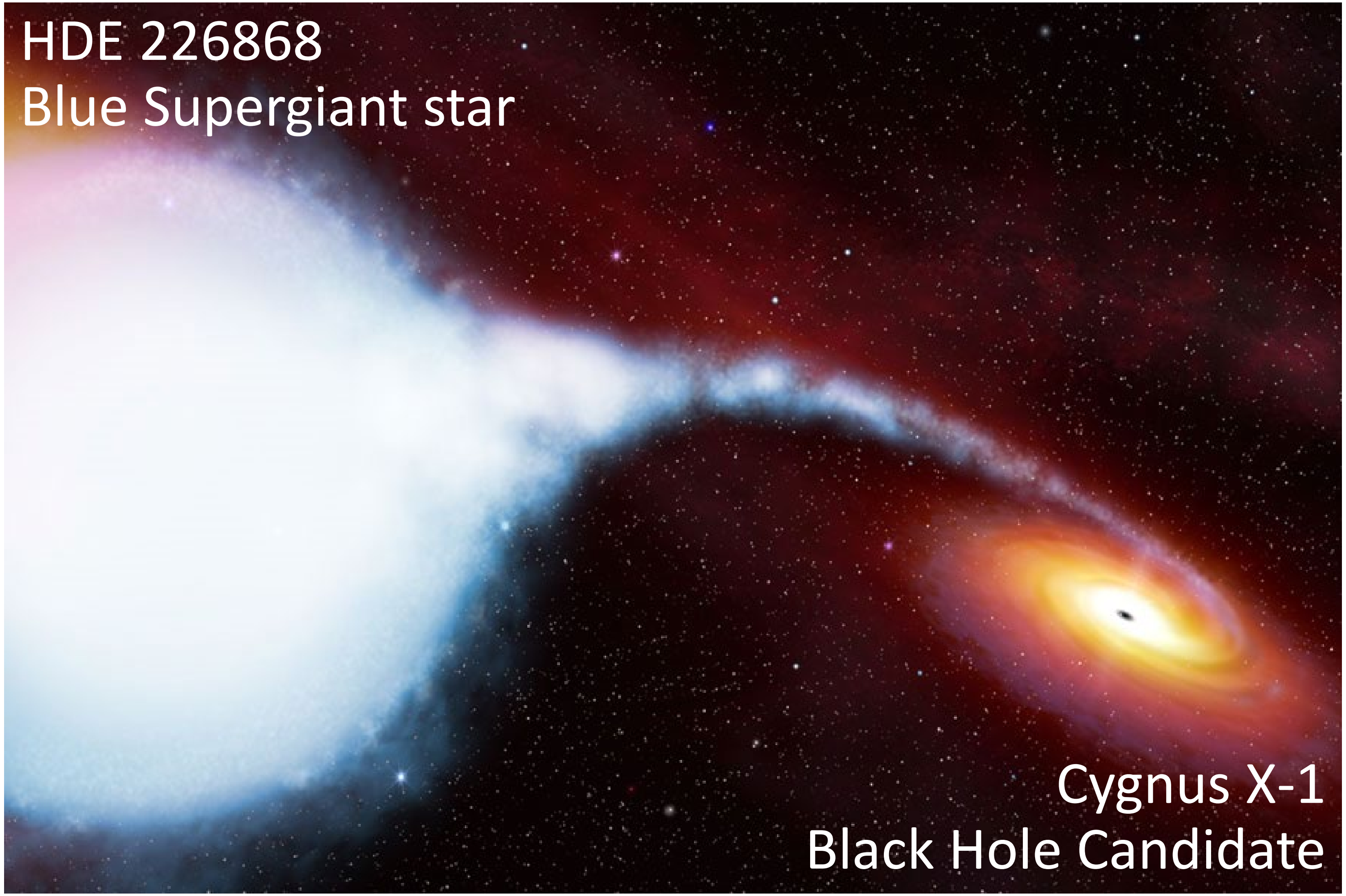






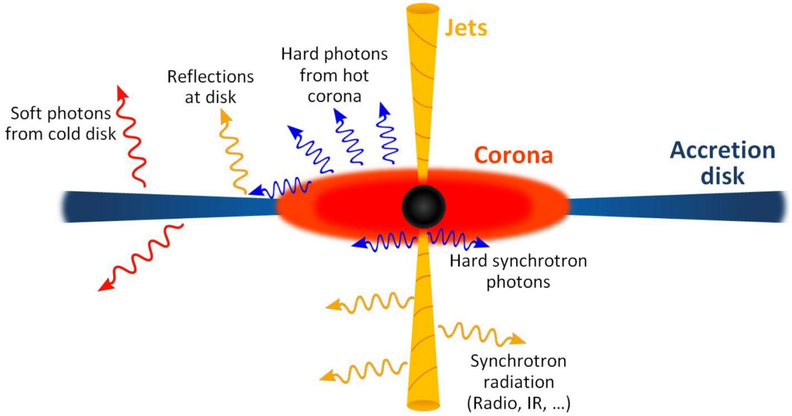


ComPol project (MPP-TUM, Munich)



Cygnus X-1 is binary star system that contains one of the best candidates for a **black hole**.

The accretion disk that is created around the black hole, fed by the stellar wind coming from HDE226868, is the **brightest persistent source of hard X-rays** in the sky.



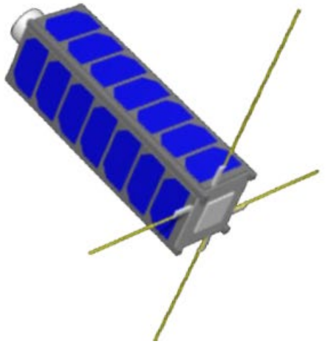
ComPol project:
Observe polarization and spectra in the hard X-ray range (20keV-2MeV)

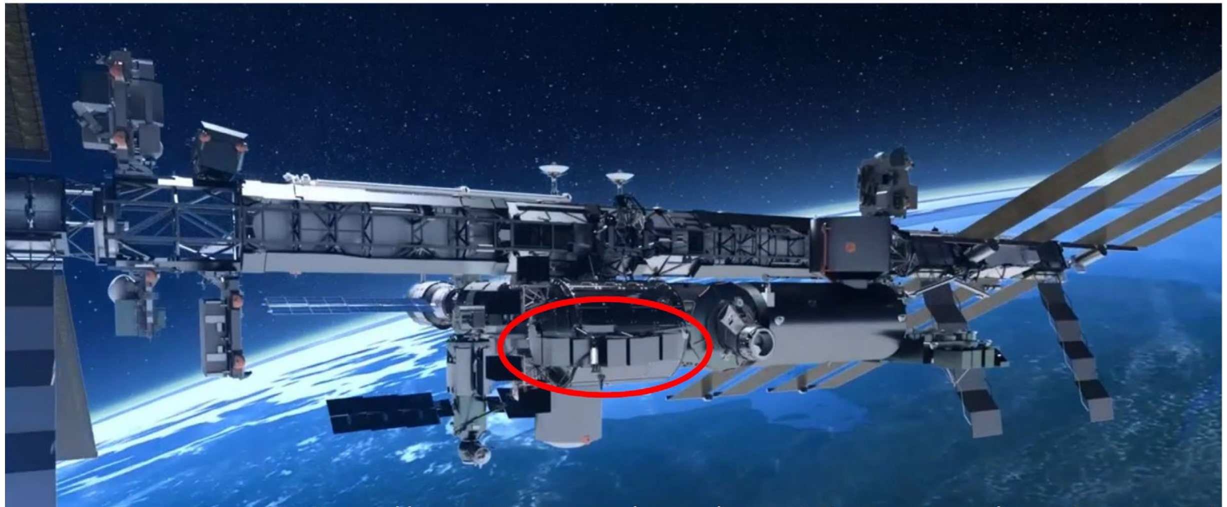
CubeSat project (MPP-TUM)

RadLab

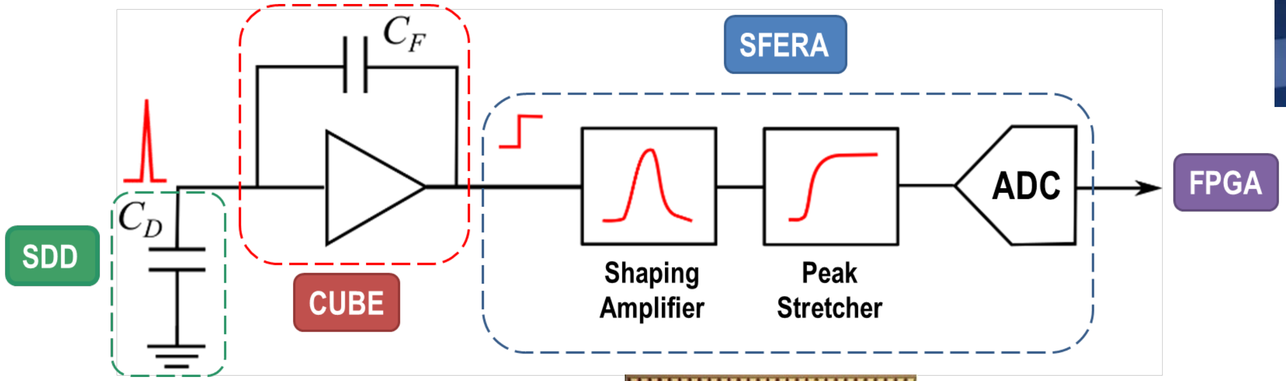
CubeSat mission

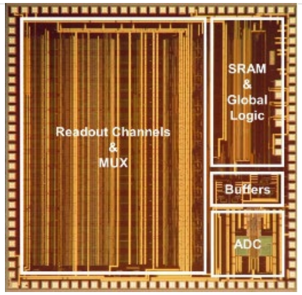
- 3U CubeSat → (10x10x34 cm^3)
- Launch date of final CubeSat >2023

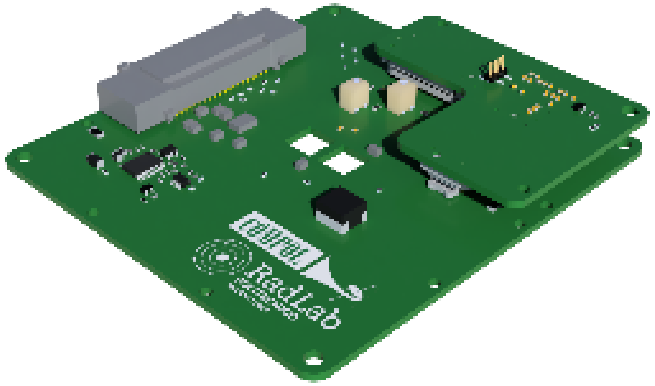


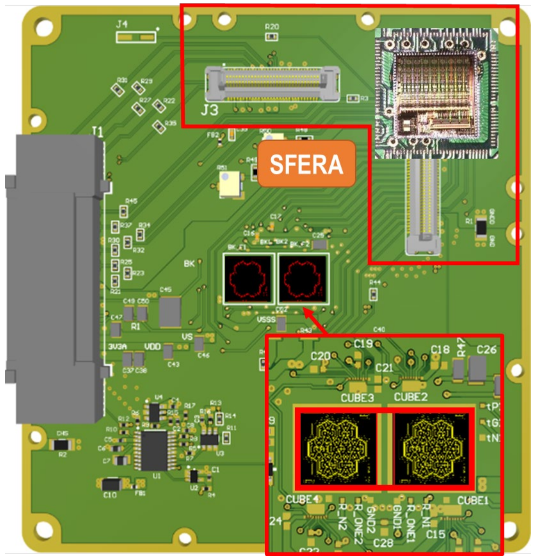


https://www.airbus.com/space/space-infrastructures/bartolomeo.html

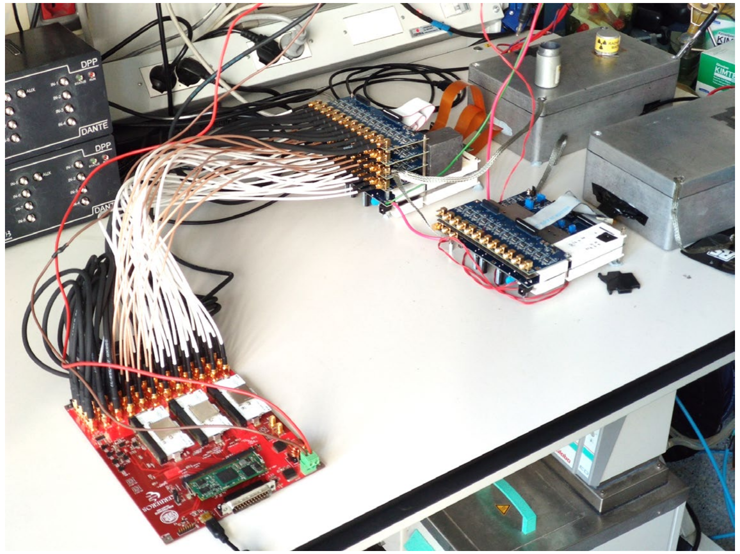


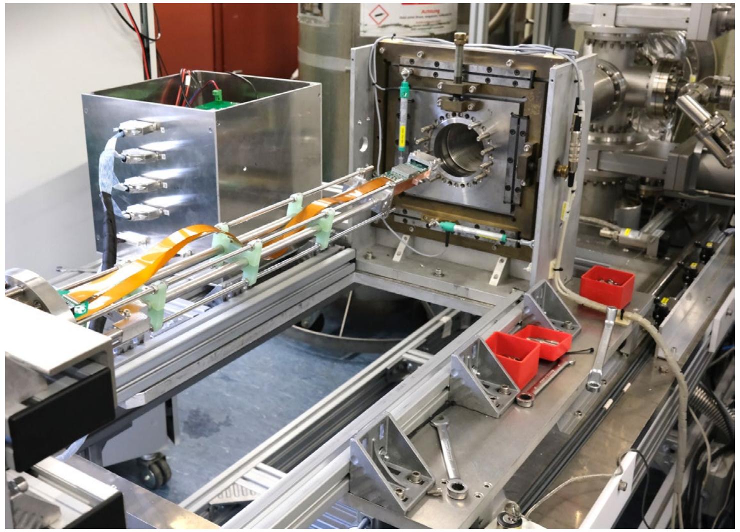






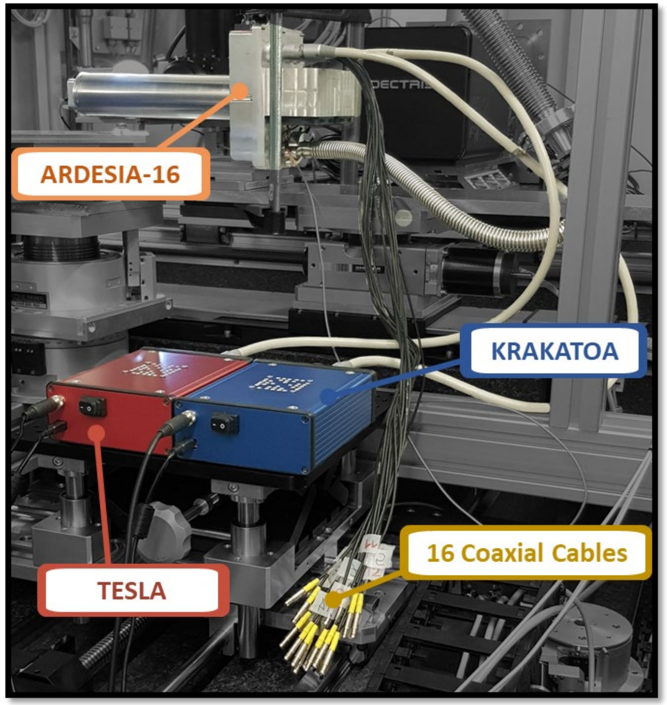
From Lab to the Experiment

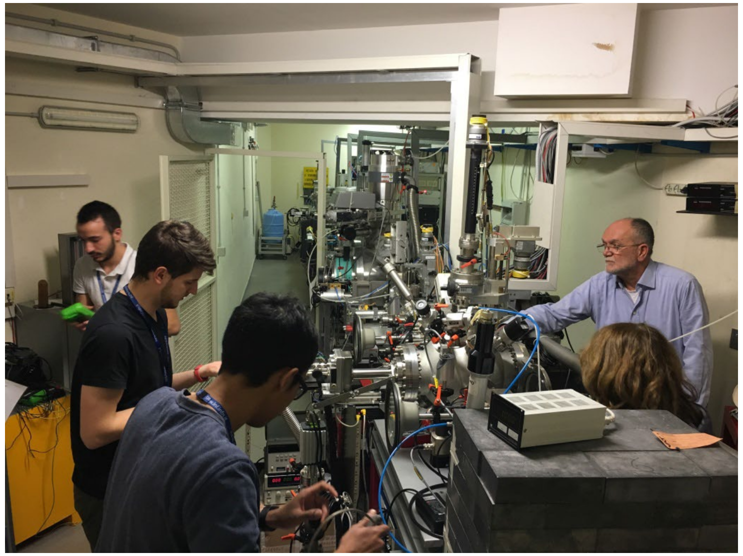


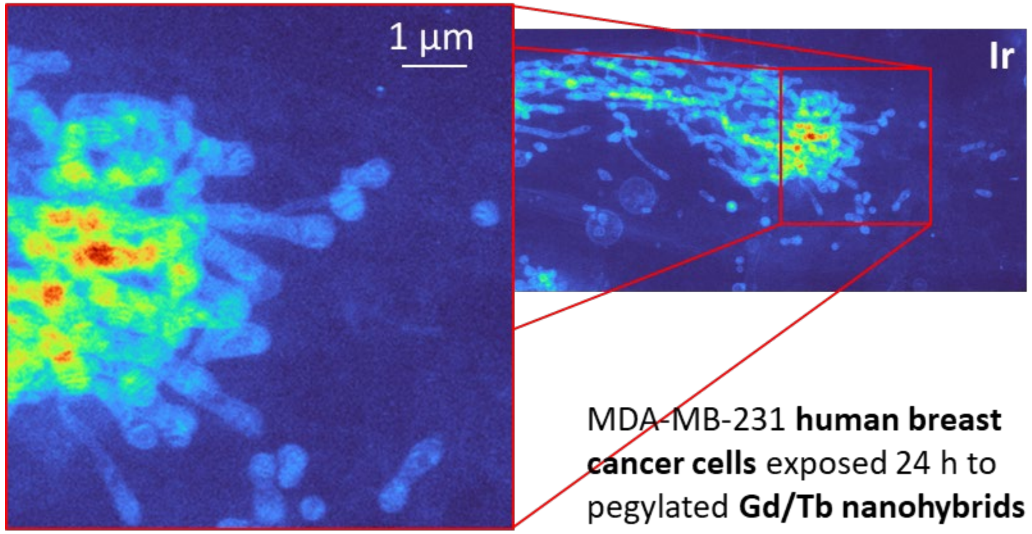


From Lab to the Experiment

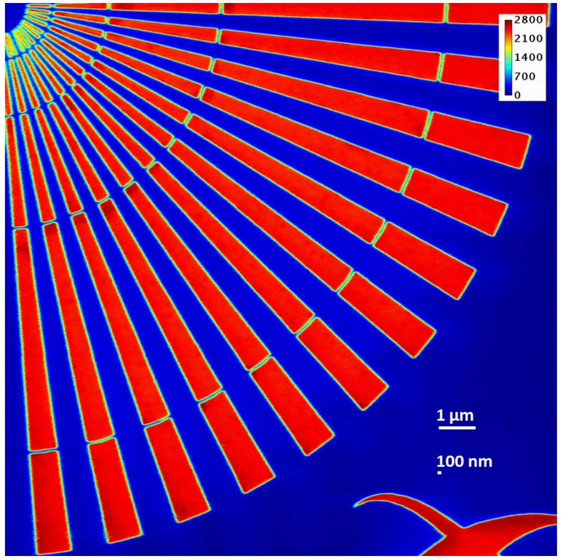
RadLab



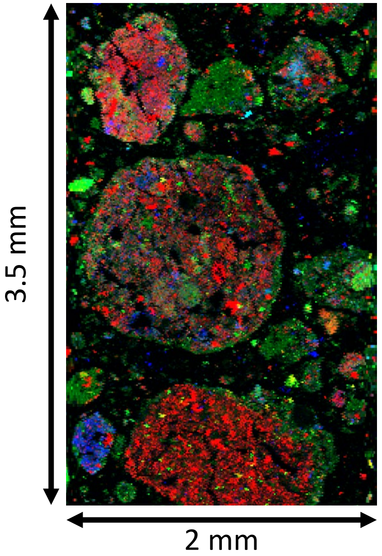




MDA-MB-231 **human breast cancer cells** exposed 24 h to pegylated **Gd/Tb nanohybrids**



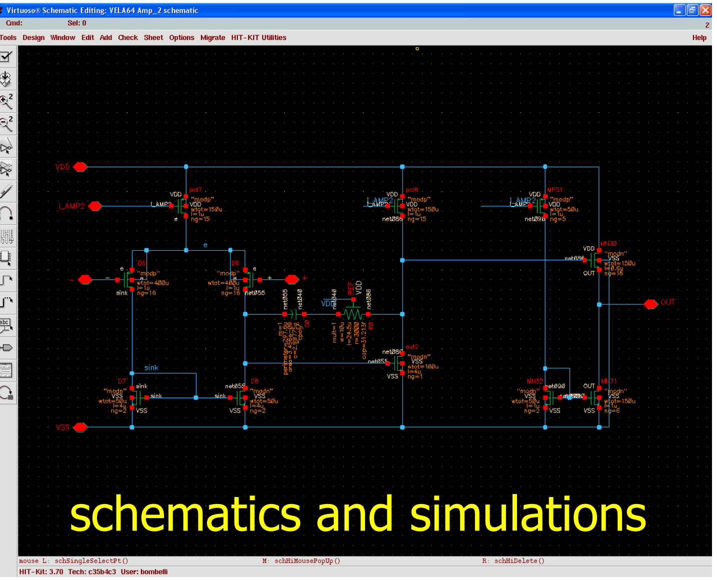
Au areal mass density (ng/mm^2)
Beam Energy: 33.5 keV
Step Size: 10 nm
Dwell Time: 15 ms
Image Size: 1500 x 1501 pixels

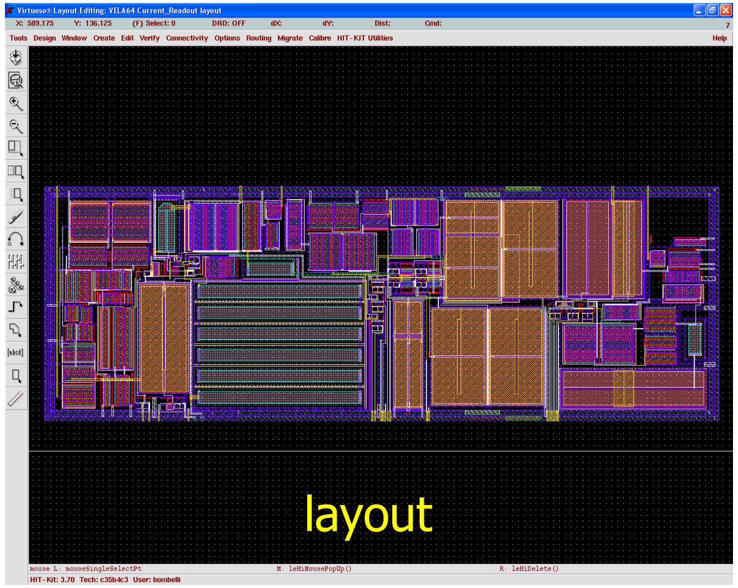


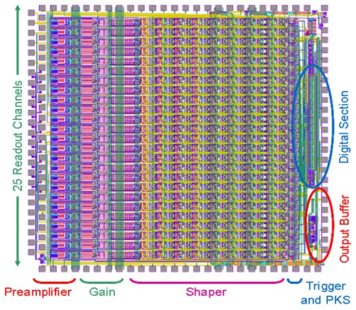
RGB image of the Cr contamination in soil. Red is Ca, green is Fe and blue is Cr.

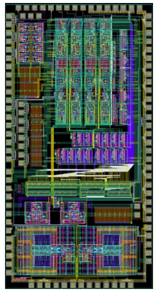
Design of integrated circuits

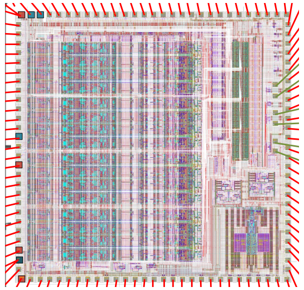
RadLab

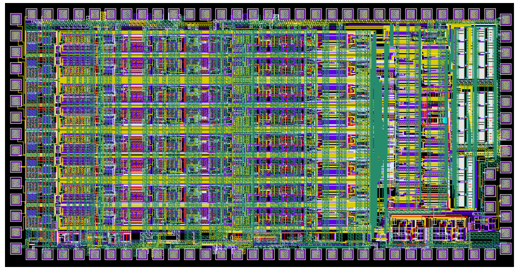


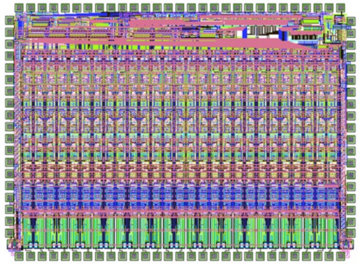


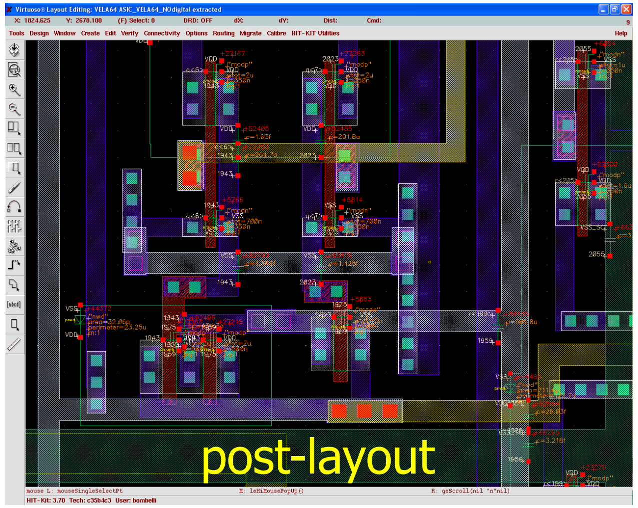


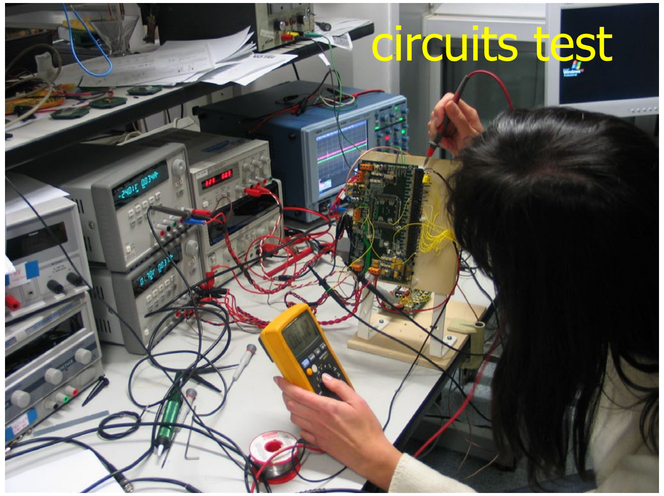


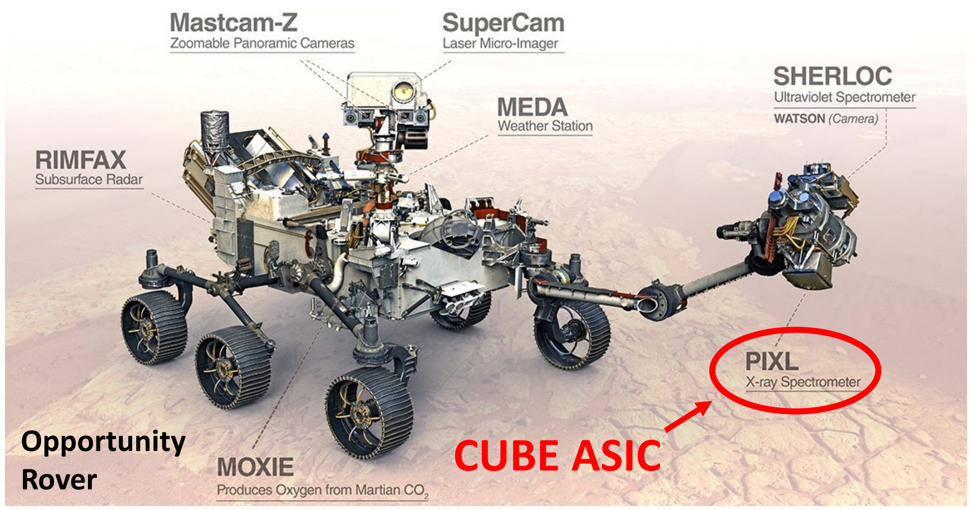






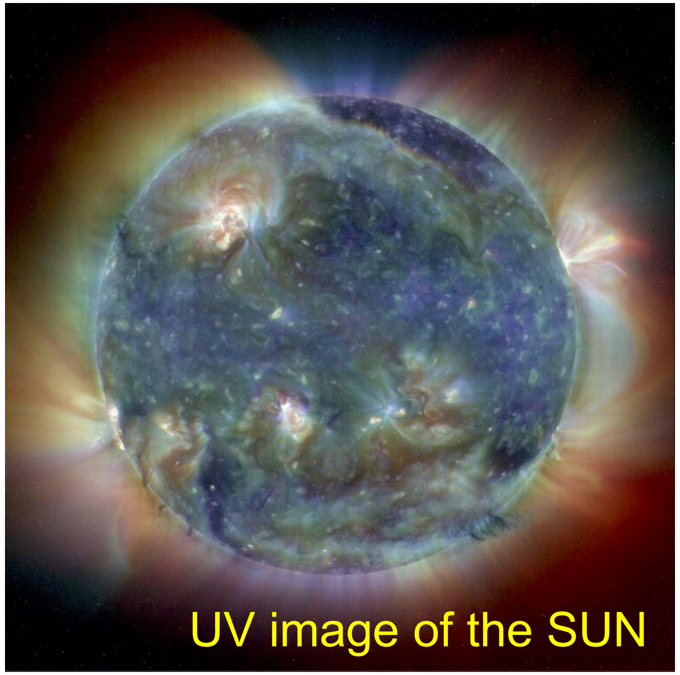


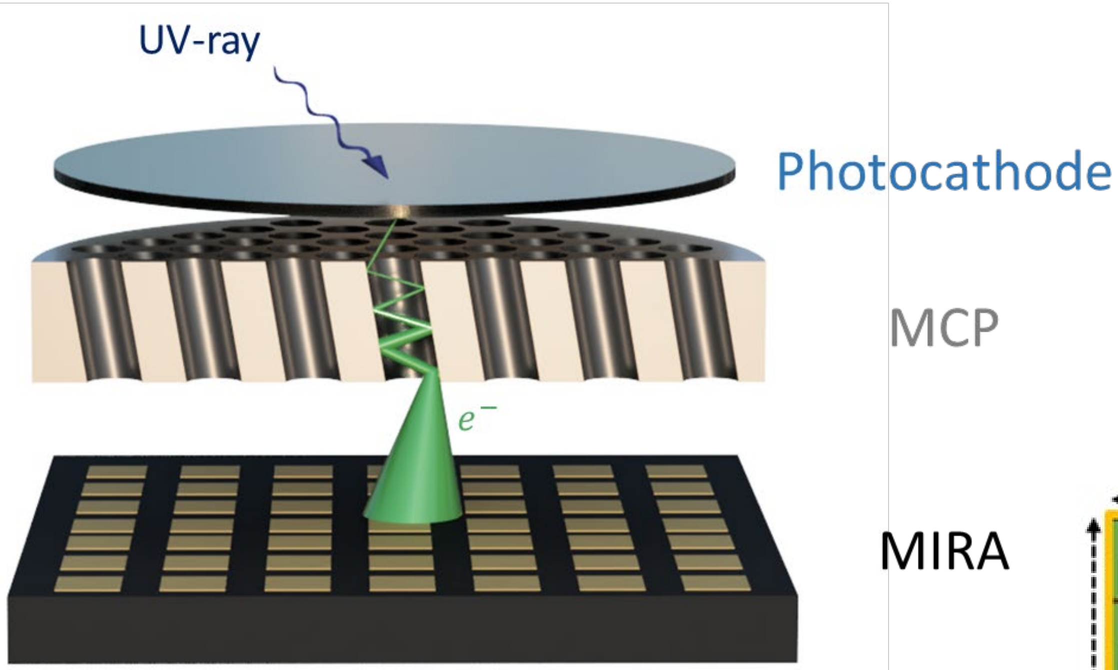




PLUS: ASIC for UV sensors for astronomy applications

RadLab





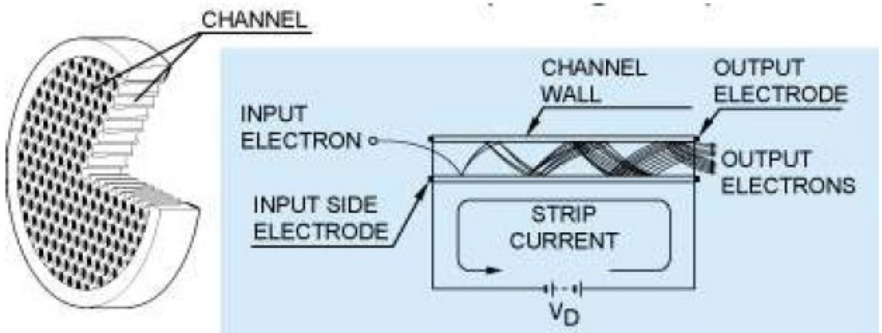
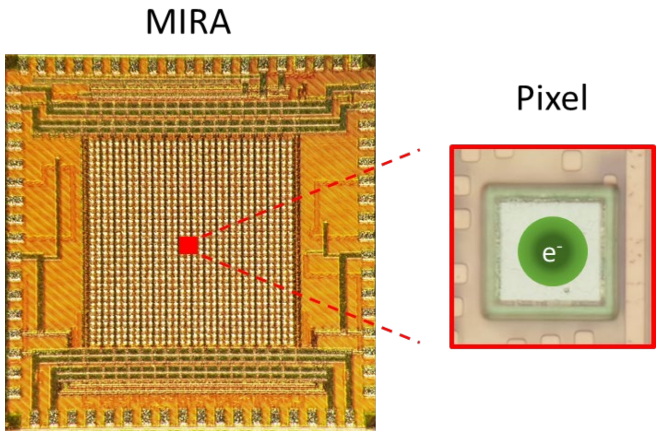
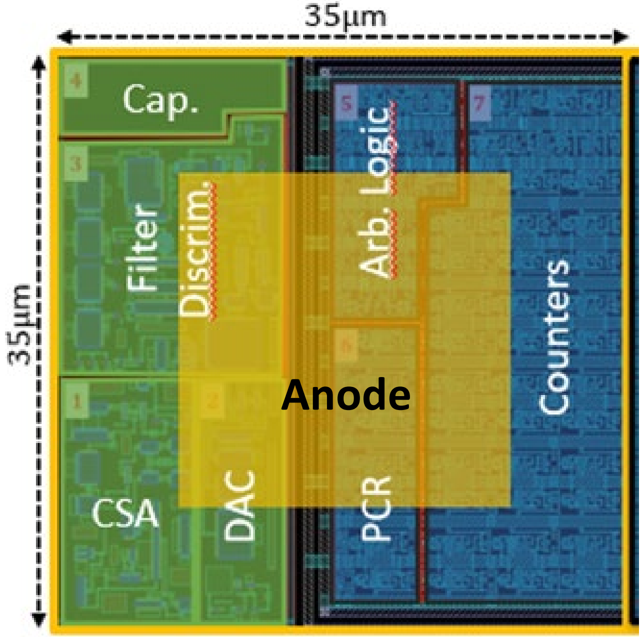


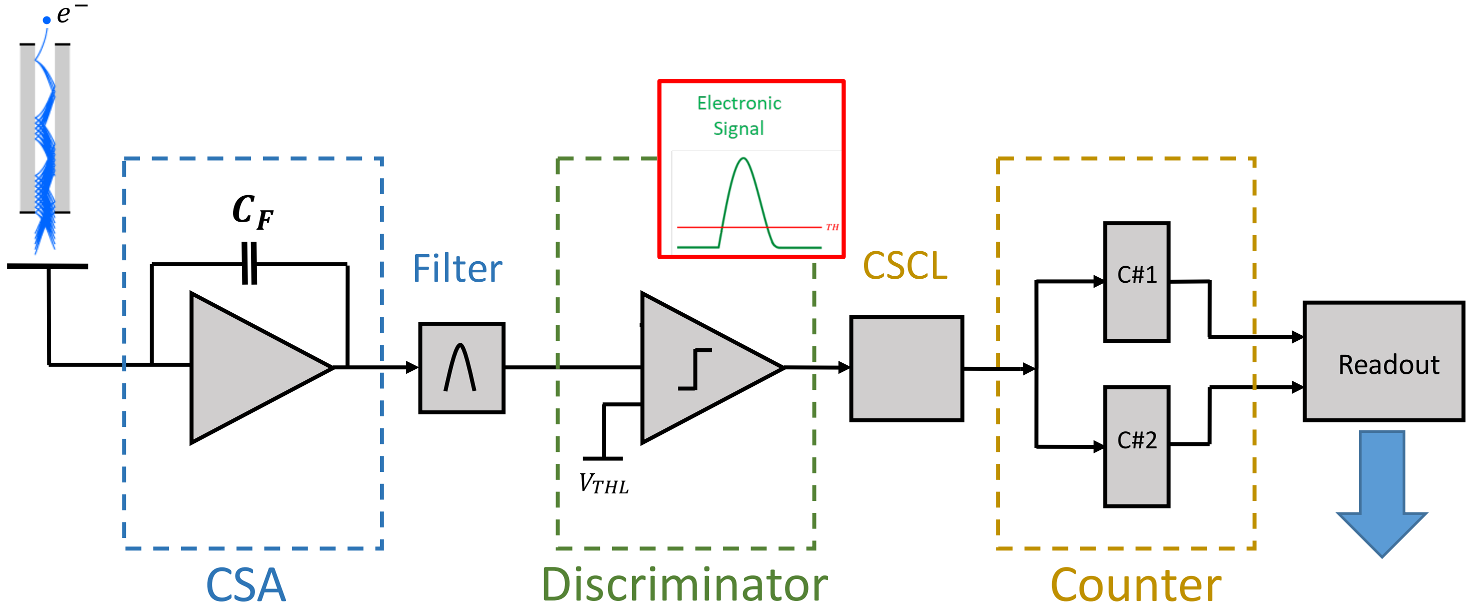
Figure 1. Schematic structure of Microchannel Plate.





MIRA ASIC Architecture

RadLab

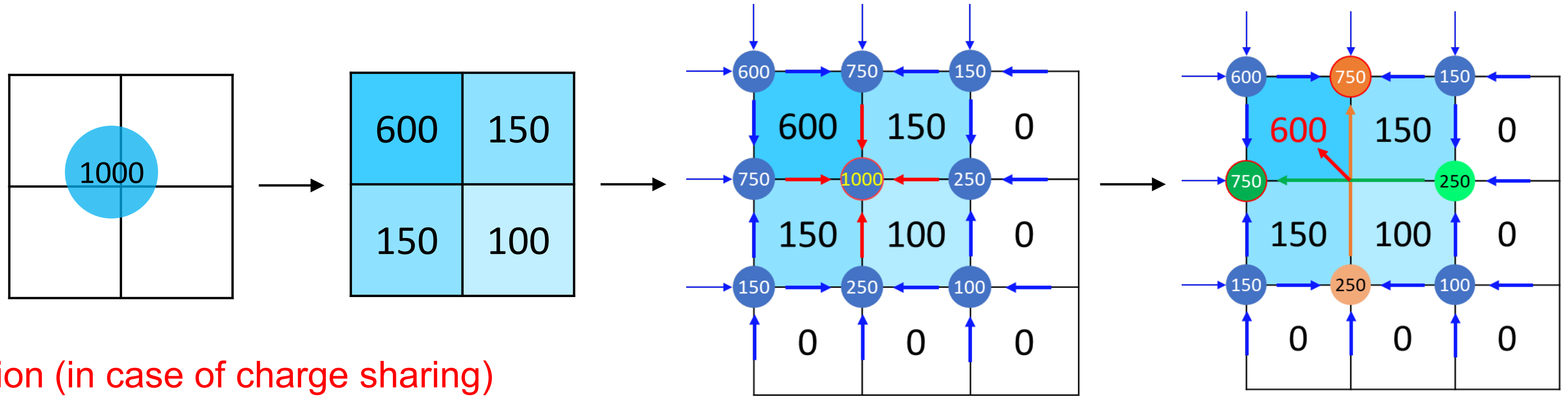


Challenges vs. state-of-the-art:

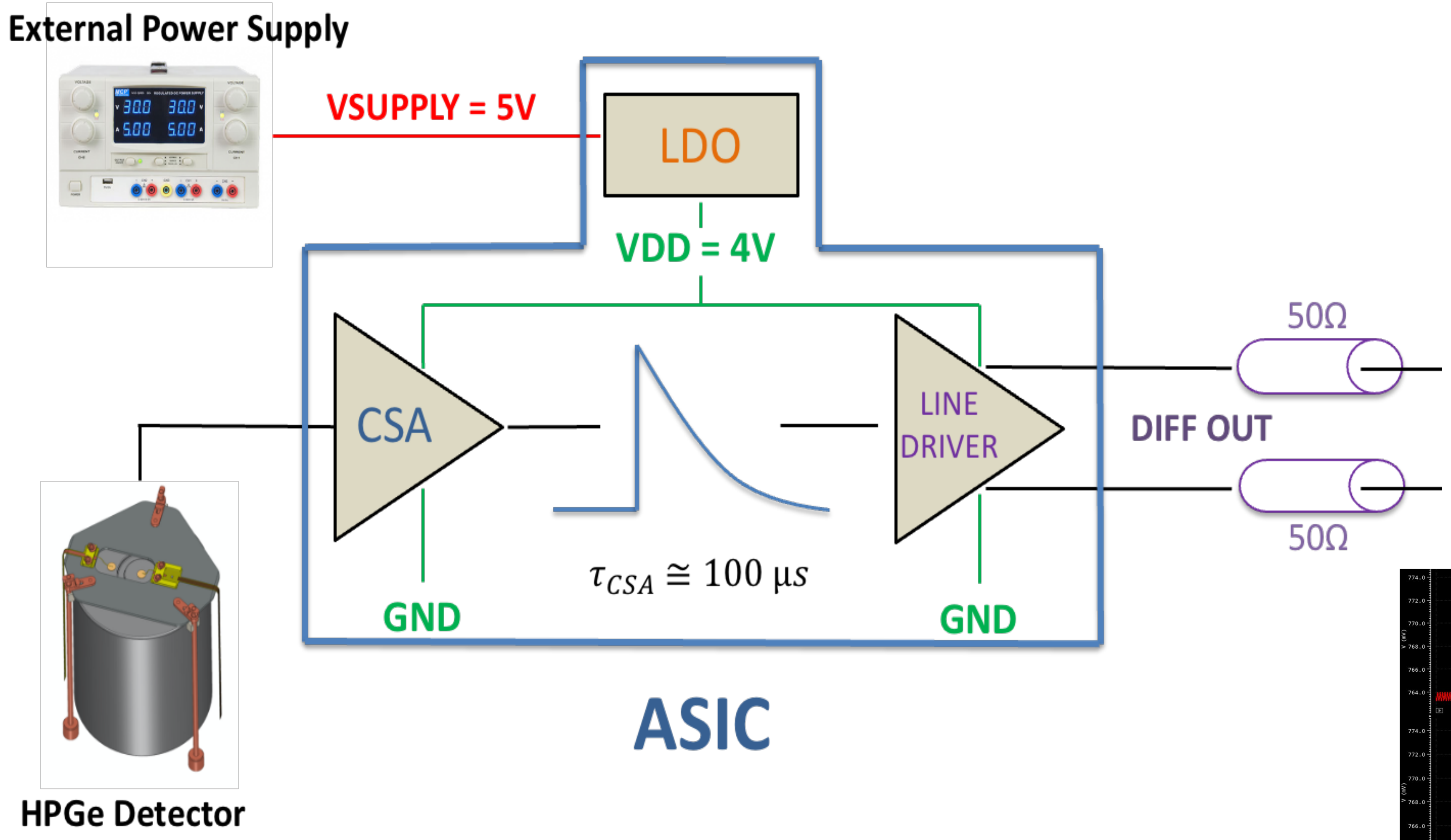
- **lower noise** (→ lower MCP gain, → higher lifetime)
- higher frame rate, lower dead-time
- small pixel size (**35um**)
- charge sharing management

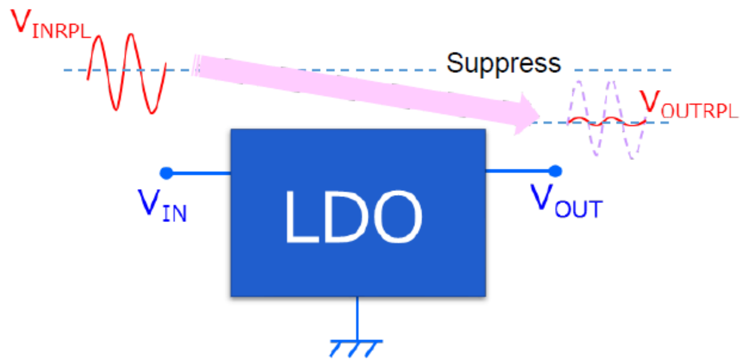
Pixel cell layout:

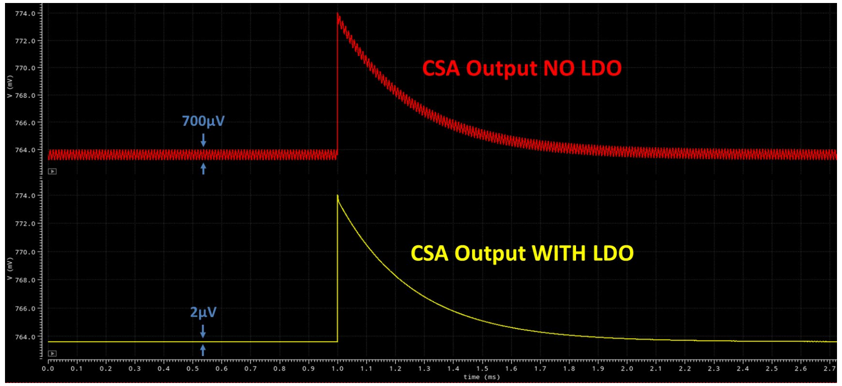
1. Preamplifier (CSA)
2. Filter
3. Discriminator
4. Control Logic
5. 2 Counters (zero dead time)
6. Configuration bits
7. Arbitration logic for event allocation (in case of charge sharing)



LEGEND 1000 Preamplifier

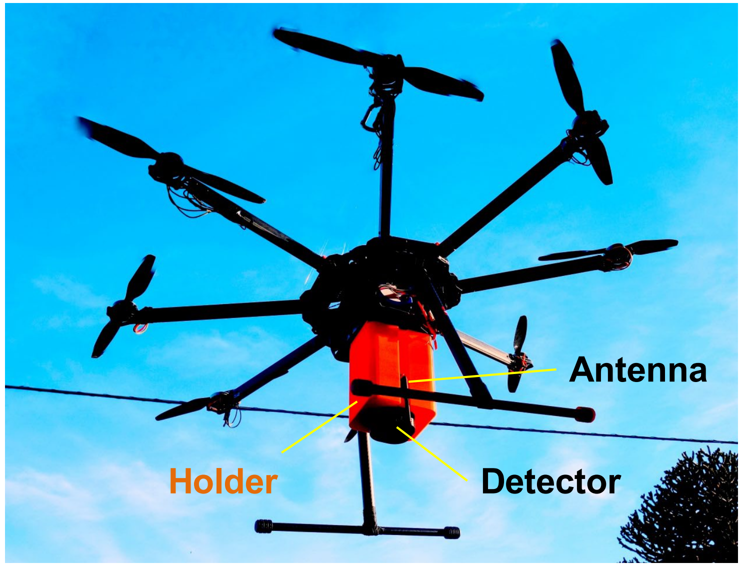


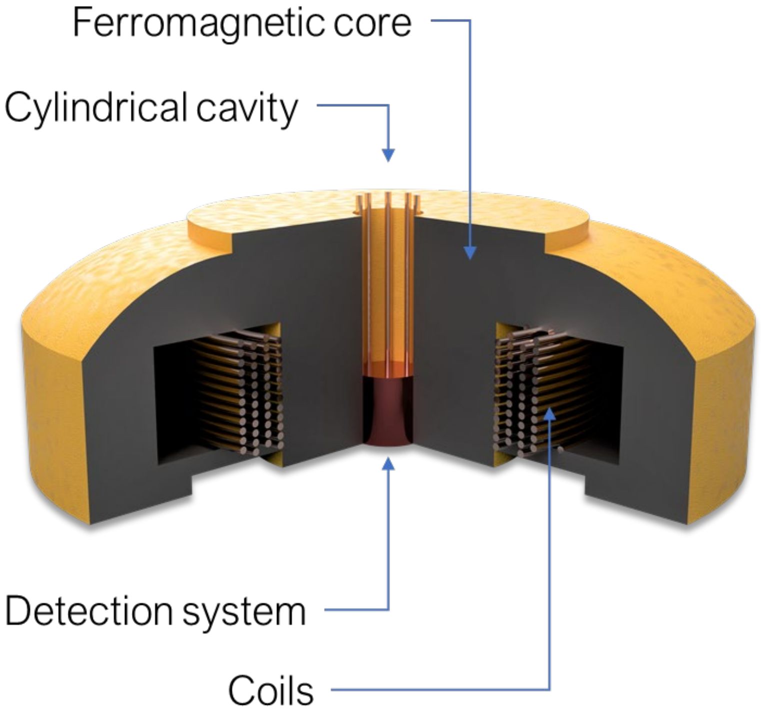




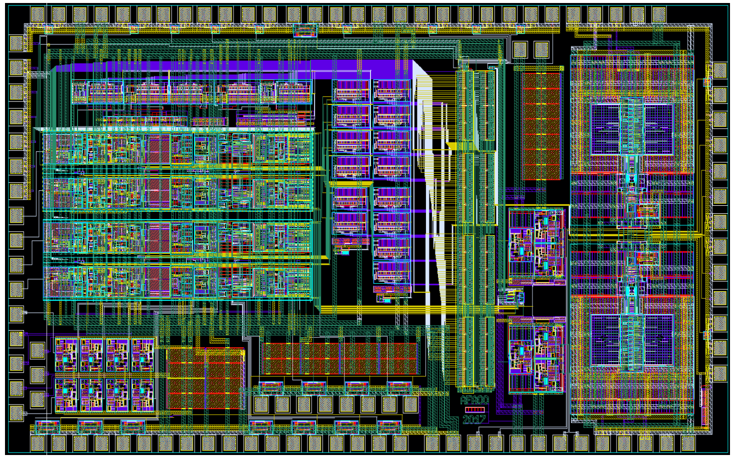
… and after the thesis, it is not over: Awards…!

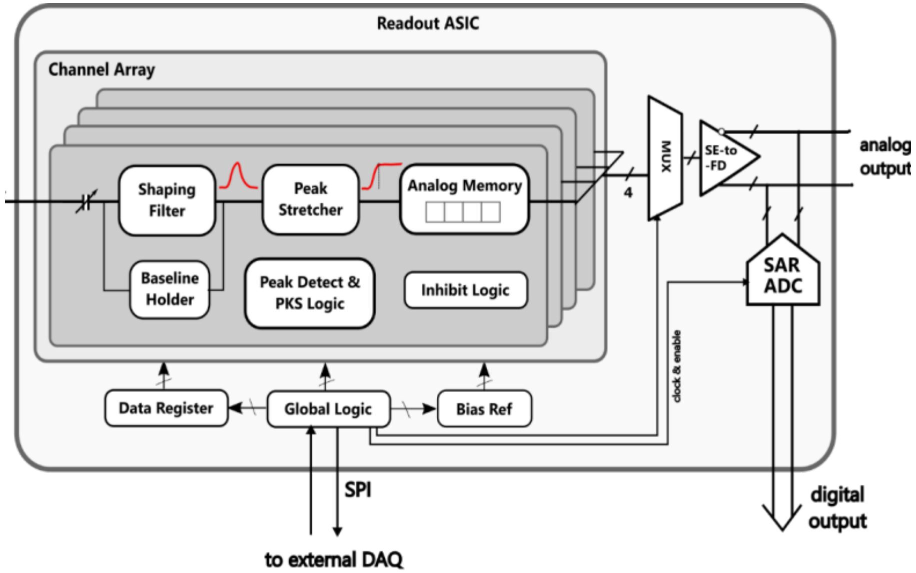
Innovation Day - Design Contest, Museo della Scienza e Tecnologia di Milano. Emanuele Lavelli, premio per miglior tesi di laurea sul lavoro: "Spettrometro di raggi gamma basato su fotorilevatori SiPM per rilevazione sorgenti radioattive".





Awards…!





Idham Hafizh, student at the Politecnico di Milano, Dipartimento di Elettronica, Informazione e Bioingegneria, has been awarded with the first edition of the Prof. Emilio Gatti Best Master Thesis Award from the Istituto Lombardo Accademia di Science e Lettere.

Preliminary list of available thesis

1) DETECTION MODULES FOR MEDICAL IMAGING
- Detectors and integrated circuits for Medical Imaging systems (SPECT/PET)
- Machine learning techniques applied to the reconstruction of the event in the detector (position, energy, timing)
- Design of an analog neural network in a CMOS integrated circuit (ANNA)

2) DETECTORS AND ELECTRONICS FOR HADRON THERAPY
- Design of a new gamma camera and related electronics for dose range monitoring in proton/carbon therapy (CNAO)
- Development of innovative detectors for Boron-Neutron-Capture-Therapy (BNCT) and their test at LENA reactor (Pavia)
- New readout ASIC (SITH) for hadron therapy applications (high energy, timing, ..) and PET
- Development of a Data Acquisition System for an industrial project in radiation therapy (TERAPET, Geneva)

3) ARDESIA/ASCANIO
- Development of a 16ch/64ch ARDESIA detector, installation and beam tests at ESRF synchrotron (France) and PETRA (Hamburg)
- Development of a new detector ASCANIO with anular configuration and related electronics

4) TRISTAN
- Development of new multi-element detector for the TRISTAN experiment and related electronics

5) COMPOL/SIDDHARTA
- Development of an SDD+electronics module for a CubSat mission on board of the ISS (International Space Station)
- Development of an SDD+electronics module for the SIDDHARTA experiment at Frascati (INFN)

…continuation

6) GAMMA

- Development and experimentation of the detector based 3'' LaBr3 scintillator for nuclear physics experiments in accelerators
- Development of imaging algorithms for Doppler broadening correction, based on machine learning and neural networks

7) PLUS

- Development and test of a new pixel readout ASIC for UV imaging in astronomy applications

8) LEGEND ASIC

- Development of an innovative low-noise CMOS preamplifier for the readout of Germanium detectors for the LEGEND-1000 neutrino experiment

9) LUXOTTICA

- Development of a miniaturized and low-power system to track the eyes

Thesis abroad (6 months):

KTH (Royal Institute of Technology in Stockholm):

- Topics: shear wave elastography, multimodality contrast agents, Finite Element Method to simulate tissue stress during accidents, …

ESRF (European Synchrotron Radiation Facility):

- Experimental characterization of XIDER, a novel X-ray detector for synchrotron applications

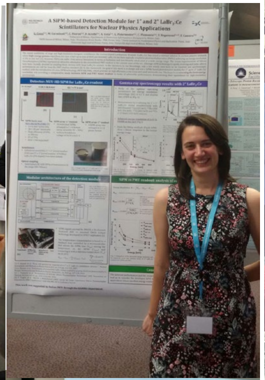
Welcome to visit the laboratory!

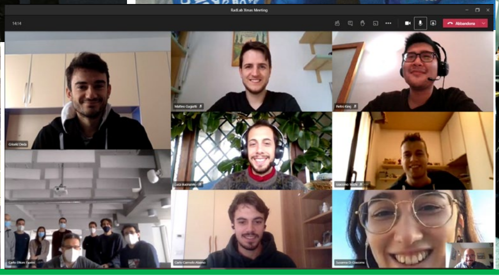
RadLab

You are very welcome to visit our lab, to talk with master students and PhD students and spend some time to see what they are doing…..











For visiting the lab and looking to research and development activities (not necessarily only for thesis interest), you can organize youself in small groups (3-4 max.) and provide me by mail desired time slot (day and time) to organize the visit.